

RESEARCH

Open Access



The impact of firm-level innovation on labor productivity and employment in selected African countries

Zerayehu Sime^{1*}  and Getaw Tadesse²

*Correspondence:
zerayehu2005@gmail.com

¹ College of Business
and Economics, Department
of Economics, Addis Ababa
University, Addis Ababa, Ethiopia

² AKADEMIYA, 2063 Kigali,
Rwanda

Abstract

The paper examines the impact of different types of innovations—such as process, product, and R&D—on labor productivity and employment across several African nations. Utilizing data from the World Bank Enterprise Survey, the study constructs pseudo-panel data based on legal status, firm size, and industry type and employs the propensity score matching and the dose–response model to explore the relationship between innovation and labor market outcomes. The descriptive analysis indicates that 2,673 firms are engaged in at least one of the innovation activities, and 250 firms do not have any innovation activities. The propensity score matching finds that the overall combined effect of all types of innovation is statistically significant and has a positive effect on the employment of non-production workers and skilled workers, but it harms their productivity. Not only this, but it has also a negative impact on the productivity of permanent, and unskilled workers. The only positive impact goes to the productivity of production workers. Except for skilled labor, no segments of the labor force have been impacted by product innovation, but it has statistically significant influences on the productivity of all types of workers, still, production workers received a positive influence. It also finds that process innovation positively influences the productivity of production workers and the employment of skilled, non-production, and production workers. On the other hand, permanent, non-production, and skilled employees respond negatively to process innovation. Regarding R&D innovation, except for production workers, all are negatively impacted and statistically significant. On employment creation, R&D has also a positive and statistically significant impact on employment except for permanent employees. On top of this, the findings of the dose–response function are also in line with the propensity score matching method except for the combined effect of innovation on production workers. In other words, the intensity has a declining dose effect while innovation has a growing effect by the propensity score method for production workers. Additionally, the dose–response function shows a strong positive correlation between the intensity of innovation and skilled workers with various trends as well as permanent employees. The effect is initially diminishing for unskilled workers and non-production workers, but it gradually increases over time. In general, the finding challenges the assumption that innovation uniformly improves productivity, instead revealing a more complex

dynamic where job creation comes at the cost of lower productivity in certain labor segments.

Keywords: Innovation, Labor productivity, Employment, Propensity score matching, Dose–response models, Firms, African countries, O31, O32

JEL Classification: O31, O32

Introduction

In literature, the role that firms play in the activities of innovation has been emphasized for many years. A variety of theories, discussions, and disagreements have been made (Atoko & Tlangelani, 2024). Firms are thought of as the catalyst for economic growth and as a gear-shifter for transforming the economy. They are essential to the operation of the market through resource allocation, production, and creation of job opportunities. Government policies, institutions, markets, and other systems all play a part in coordinating these transactions. Numerous theories and frameworks have been developed to explain how firms make decisions and behave to maximize profits (Foss, 1998). The creation of new technologies and techniques targeted at raising productivity will reflect this. Economies would grow more slowly without this innovation and use of capital. Manufacturing firms in this regard play pivotal roles and generate multiple effects on the other economic sectors and lead to economic transformation in a non-linear pattern. This dynamic process continues until manufacturing takes the lead position and contributes to GDP (Chenery & Taylor, 1968; Kongsamut et al., 1997; Kuznets, 1966).

Innovation has recently been acknowledged as a key component of the green industrial revolution that determines productivity growth and sustainability. It is audaciously included in the Sustainable Development Goal (SDG), which acknowledges its significance and ability to enable research and development efforts that achieve all other SDGs (Fendoğlu & Polat, 2021). It is also believed that innovation creates a significant, pervasive, and long-lasting disparity in business productivity even in developed countries (Foster et al., 2001; Jorgenson, 2011; Syverson, 2011). For instance, extensive innovation was a major factor in the rapid rise of Asian economies in the 1970s and 1980s by securing durable and sustainable productivity growth (Bloom et al., 2019; Fagerberg & Verspagen, 2002; Fagerberg et al., 2010). As a result of this, consumers benefit from innovation by receiving lower prices and higher-quality items, diverse goods and services (Fagerberg, 2006; Simanis & Hart, 2011). However, the empirical evidence revealed that the nexus between innovation, productivity, and employment is complicated (Edquist et al., 2001; Harrison et al., 2014).

African governments in this regard have tried to implement several reforms and reconsider their industrialization policies. In addition, they provide certain industries with extra oversight and support, expand access to development resources, and enforce laws requiring joint ventures with international companies. The incentive packages also contain cash incentives, income tax, and customs. The objective of these packages, despite their varying techniques and conditions, is to assist businesses and facilitate the nation's industrialization. (EIC, 2020). Since the year 2000, the continent's economic growth has been steadily dropping, going from a peak of 6.6% in 2002 to – 2% in 2020 (WB, 2019). When research and innovation data are considered, the situation becomes worse. Africa

generates 0.1% of all patents, 1.3% of research spending, and barely 2% of global research output (Simpkin et al., 2019). In all areas of innovation, the participation rate is only 10%. Only 0.1 percent of GDP is spent on research and development (UNESCO, 2018). These put a daunting challenge on labor productivity and employment opportunities. Therefore, the study raises the following research questions: (1) what makes up productivity, employment, and innovation at the firm level in Africa? (2) Do different types of firm-level innovation affect different labor groups and their productivities in the same ways? and (3) Do the effects of innovation intensity and selection differ from one another?

Despite the significance of this issue, only a small number of studies, meanwhile, looked empirically into the relationship between employment, productivity, and innovation (Getaw et al., 2022). Based on statistics from Africa, a more thorough examination of the intricate and complex relationship between innovation, employment, and production is required. Robust analysis techniques such as dose–response modeling and propensity score matching was not used in most previous studies to ensure robustness. Additionally, they were unable to fully capture the dynamic of innovation by looking at various labor segments and kinds of invention as well as by including lag time in the model specification. This study's contribution is to close these research gaps. The issue of discovering a novel approach to capturing the intersection of firm-level innovation, employment, and productivity is important to industrial economics and labor markets, which underlines the significance of this research in the African context.

The primary objective of this study is to assess how various forms of firm-level innovation influence both productivity and employment in different segments of the labor force (e.g., skilled, unskilled, production, and non-production workers). Additionally, the study aims to determine the extent to which innovation intensity—measured by the “dose” of innovation—affects these labor outcomes using pseudo-panel data from the World Bank Industry Survey.

By studying the effects of firm-level innovation on labor market dynamics and productivity in Africa, this study adds to the corpus of knowledge on the subject. The study's potential to assist governments, international organizations, and business communities in developing policies and strategies that support various forms of innovation and labor segments accounts for its practical value. These findings are essential for ensuring productivity and employment-creation benefits of innovation. By doing this, the study's originality is upheld and demonstrated as rigorous scientific research is carried out; all sources and support obtained throughout the creation of this work have been acknowledged; the research gaps are filled; and certain values are added in the context of Africa.

Review of related literature

Theoretical literature review

The theory of innovation is traced back to the first work of Joseph Schumpeter during the pre-1950 period. He was also the first to use the term “creative destruction” to describe how the old is being constantly replaced by the new (new technologies, new kinds of products, new methods of production, and new means of distribution) and identified the most prominent and linear stages of innovation (Schumpeter, 1934). Later on, the two versions of the linear model of innovation (technology push and market pull) emerged in the 1950s and up to the early 1970s. However, they were criticized as they

gave more emphasis on incremental change in innovation and neglected a range of innovation dimensions including its complex nature (Foxon, 2003; Nemet, 2007; Philip et al., 2012). Induced innovation, evolutionary approaches, and path-dependent models then emerged in the 1970s to 1990s (Ruttan, 2001). The paradigm that was shifted to reflect the complexity and interdependency of the innovation process over the period 1980s to the 2000s is known as the innovation system as designed by the Oslo Manual (OECD, 2005). Thus, following the World Bank Enterprise Survey, this study categorized innovation into three: (1) product innovation comprises adopting new or improved goods or services, introducing them to the market, and developing new or better ways to manufacture goods or render services; (2) process innovation focusses on developing new or better organizational structures, new or better marketing strategies, new or better management practices, and new or better logistics, delivery, or distribution systems; (3) R&D innovation looks at employee time spent developing new ideas, and R&D expenditures.

Both innovation and productivity are closely intertwined in the firms' optimization process through an effective way of doing business, lowering the cost of production, optimizing processes, and providing a guarantee for market relevance. Utilizing innovation-driven scale economies increases the production of goods and services relative to inputs, leading to a positive relationship between innovation and productivity. The other explanation indicates that innovation enables firms to exhaustively run unused capacity utilization, and thus increases outputs without hiring additional inputs. On top of this, innovations have led to new methods of creating previously known goods that require less input, and thus the production frontier has shifted outward, enhancing efficiency. As a result, firms can boost output, which raises overall productivity in general (Farrell, 1957; Pierre & Hall, 2013).

Specific to product innovation, there are two scenarios depending on whether the newly introduced products replace the old or it added to an established business. As the new products replace the old products and push them out of the market, the revenues from producing the old products may be cannibalized since they require fewer inputs to produce than the old products. So, a new product when introduced to the market, generates new demand, which can lead to scale economies in its production or increased productivity. In the case of adding a new product, the opposite may occur. Economies of scale of the new products concurrently with the old ones. The firm possibly progresses through the learning curve, productivity initially drops and then increases later on.

On the other hand, process innovation allows new processes to remove more expensive factors of production, which lowers production costs, and enhances productivity. Additionally, R&D innovation supports firms' absorptive capacity (ability to identify, assimilate, transform, and use external knowledge), which is necessary for learning by doing. Firms that engage in internal research and development have access to external knowledge, promote spillovers, and increase productivity (Cohen & Levinthal, 1989; Griffith et al., 2004; Pierre & Hall, 2013). Besides, it has also an indirect effect on the reaction of the market to innovation. For instance, a productivity-driven price decrease, and if demand is price-responsive enough, leads to a more than proportional increase in sales and this increase in sales can then spur additional productivity increases. The degree of market rivalry, which can be a consequence of

the innovation's relevance, impacts how much the unit cost reductions translate into a price decrease (Pierre & Hall, 2013).

The introduction of new or significantly enhanced products should increase demand for innovative firms and, consequently, employment levels. Nonetheless, innovative firms that have a temporary period of market dominance could establish pricing that maximizes profits and cut production to the point where the net effect on employment following the substitution of the new offering is negative. However, even though process innovation usually results in labor savings, its impact on employment is not always clear-cut. If fewer people are needed to produce the same amount of labor, the company can pass this efficiency advantage on to customers in the form of lower pricing, which will boost demand. Compensation mechanisms can offset the labor-saving impact of process innovation, depending on market structure, demand elasticity, and the elasticity of labor-capital substitution (Spiezia & Vivarelli, 2002).

The intricate link between employment and innovation hinges on whether innovation creates or destroys jobs, and could be the main source of employment dynamics (Blanchflower & Burgess, 1996). The impact of innovation has also quality and quantity aspects. The quality aspect of the impact examines the type of employment that innovation either creates or destroys, leading to changes in skill changes and wage structure. While the immediate effects of research expenditures that result in process innovation are typically labor-saving, spending on R&D that leads to product innovation is typically labor-friendly (Vivarelli, 2015). Product innovation may imply the emergence of new firms and new sectors and thus new jobs. The main point of contention in the present product innovation debate is whether or not innovation entails job creation. The impact of product innovation on labor-friendly conditions is indisputable, yet it might vary depending on the situation. The "substitution effect", which refers to the replacement of mature products with new ones, must be weighed against the "welfare effect" of product innovation, which is the development of new goods (Katsoulacos, 1986).

Process innovation leads directly to labor-saving (job-destroying) effects. This is primarily because of the introduction of labor-saving innovation that enables the production of the same amount of output with a reduced workforce. Through the introduction of new goods and markets, product innovation may have the impact of generating jobs. However, even in the case of process innovation, market processes ultimately determine the influence on labor demand, they can offset and compensate for the direct impact of job destruction through new machinery, lower prices, new investments, and lower wages, among other factors (Vivarelli, 2015). Though process innovations create job displacement, they create additional labor demand in the upstream industries that produce the new machinery. The net effect of such a mechanism is vague depending on the efficiency of their relative cost of labor and new machinery. Though process innovations result in employment losses, they also bring about changes that lower average costs. This effect is translated into reduced prices under the assumption of perfect competition, which implies greater demand and hence more production and jobs. However, this line of reasoning breaks down if there are demand rigidities, as untranslated accumulated profit is not converted into new investment (Vivarelli, 1995; Vivarelli, 2007; Vivarelli, 2014; Freeman & Soete, 1987).

Empirical evidence

There is conflicting empirical evidence on the impact of product, process, and R&D innovation on different segments of employment and their productivity. For instance, Chudnovsky et al. (2006) and Masso and Vahter (2008) found a positive effect of process innovation on productivity. However, Janz et al. (2004) and Van Leeuwen and Klomp (2006a, 2006b) found the opposite. Hall's (2011) result was a bit different and ambiguous relationship between process innovation and productivity, but there was no doubt about the positive effect of product innovation on productivity when he employed the CDM (Crepon, Duguet, and Mairesse) model and measured the real revenue effect. According to him, the ambiguity stems from our inability to quantify process innovation's true quantitative effect—the quantity that matters most for social welfare. Pierre and Hall (2013) have researched the same issues based on the CDM model to confirm a positive relationship between all types of innovation and productivity. But, they held that the impact of innovation varied depending on the production itself and the price at which it is sold. It is challenging to distinguish between these two effects in the lack of reliable individual pricing measurements. Isolating the distinct effects of each form of innovation is challenging due to the imprecise assessment of innovation and its simultaneity. While pinpointing the precise point of complementarity between them is difficult, there appears to be some complementarity between them. Mairesse and Robin (2009) examine the effect of process, product, and R&D innovations in France through the use of a three-stage econometric model. For both observation periods, we find that worker productivity in the manufacturing industry is greatly and positively enhanced by product innovation. Labor productivity is positively impacted by process innovation as well, although this impact is greater when process innovation is combined with product innovation. Crepon et al. (1998) also concluded the same results about R&D with the demand pull and technology push indicators.

Recently, studies have appeared to compare different types of innovation as they are presumed to have different effects on productivity. For instance, Mohnen and Hall (2013) have discussed product innovation and found that it bears a stock of benefit over productivity depending on the degree of innovation to the market and/or to the firm. When it is new to the market that the firm is engaged in, it may have a higher benefit in terms of enhancing productivity as it addresses new demand in the market. This may reduce the unit cost and take advantage of economies of scale as a result of product innovation. Following the introduction of a new product to the market, it hurts the old product that existed in the market, so the driving effect of the new one may have a cannibalizing impact on the old one. As a result, there is a possibility of declining productivity at the initial stage and it may have a positive effect after a certain level in response to learning by using effect. On the other hand, Mairesse et al. (2005) and Hall et al. (2009) found a positive innovation effect on productivity, whereas Raffo et al. (2008) found a negative effect of innovation on productivity. Such inconsistency may arise because of the cannibalizing effect and learning effect.

Other than product innovation, some researchers were also interested in examining the effect of process innovation on productivity. Mohnen and Hall (2013) and Masso and Vahter (2008) suggest that process innovation has a positive effect since it is highly correlated with reducing the cost of transactions as expected. However, there is also a

possibility of having a negative effect when it has a disruption effect at the beginning due to inefficient production (Roper et al., 2008). On the other hand, Hall (2011) suggested that firms may prefer to perform at their inelastic demand and obtain market power so that they can follow a cost-centered approach and become more cost-efficient. As a result, firms' sales revenue may be declining (Van Leeuwen and Klomp, 2006a, 2006b). In summary, the empirical findings revealed inconsistent and ambiguous findings regarding the effect of process innovation on productivity. Regarding organization and marketing innovation, it is a prerequisite stage to the success of other innovations and it is a strategic move as an intangible asset. Armbruster et al. (2008) and Greenan (2003) found that it has a positive effect on firm performance as it enhances the quality, competitiveness, and flexibility of firms' procedures (Goldman et al., 1995). By using spending on research and development as a traditional proxy, the empirical studies by Mairesse, and Sassenou (1991) and Mairesse and Mohnen (2010) attempted to examine the impact of innovation on firm-level productivity from the input side and discovered that innovation had a positive effect on productivity. Research on the production side of innovation has also been done by Balasubramanian and Sivadasan (2011) and Bloom and Van Reenen (2002), who discovered that patent rights—a proxy for innovation—have a major impact on a firm's success.

Using the World Bank Enterprise Survey in Latin America and the Caribbean, Crespi et al. (2016) found strong evidence regarding the relationships between innovation output (product innovation) and productivity as well as between innovation input (R&D) and output (innovation). Additionally, they discovered that process innovation yields smaller private returns than product innovation. They also discovered evidence that spillovers from product innovation are more robust than those from process innovation. A Generalized Structural Equation Model (GSEM) is used to estimate the causal links while accounting for endogeneity. Hanan and Amira (2020) recently found that innovation is the key determinant of productivity in all countries, but the evidence is much stronger for product innovation by African firms in comparison with other emerging and developing economies. With a different method of measuring productivity, Naveed and Wang (2023) used a global sample for a panel of 87 countries from 1996 to 2013 with treatments for endogeneity and discovered that innovation had a positive significant effect on both structural change-weighted productivity and unweighted average productivity, with the effect on structural change-weighted productivity being larger. Additionally, we discover that this influence varies depending on the income level and structural transition phase of the economy.

On the same note, the impact of different types of innovation on employment has been also debatable and generated conflicting results. The influence of innovations on employment has long been a topic of discussion. Researchers have attempted to separate the impacts of innovation on compensation and displacement, but it is particularly challenging to identify the displacement and compensation effects empirically. The argument provided by Vivarelli (2015) addresses how technological innovation affects the economy through both process and product innovation, both of which can have employment impacts. For the most part, R&D expenditures that result in product innovation are generally labor-friendly, creating new jobs, while embodied technological change that results in process innovation is generally job-destroying. Innovation in products

and processes is frequently linked, and job elimination is not always the result of process innovation. Process innovation's ultimate effect on labor demand is determined by market mechanisms that, if institutional and market rigidities do not stand in their way, can offset the direct impact of job destruction. Compensation strategies based on price and income have the potential to offset the first worker displacement that follows process innovation.

Rethinking previous research that indicated a positive impact of product innovation and a lack of a clear pattern for process innovation in developing nations, Doris (2016) discovered that process innovation has an employment-generating effect in Chile, Argentina, and Costa Rica because it may be driven by innovative firms taking market share out of non-innovative firms. According to the Okumu et al. (2019) study, pooled OLS technique data from Africa likewise showed a favorable correlation between job growth and process and product innovation. Improvements in processes and products boost worker productivity without directly affecting employment. However, the rise in sales more than offsets this direct and negative impact, meaning that innovation has a net positive impact on employment if the opposing impacts are eliminated. Similarly, Cirera et al., (2016) looked into how product innovation affects labor intensity and generates significant new sales, especially in Africa where businesses are distant from the cutting edge of technology. They also discovered that as a result of production automation, process innovation has no short-term detrimental effects on business employment. Additionally, they pointed out that as it increases efficiency and decreases the elasticity of new sales to employment, evidence of a detrimental impact on employment is starting to emerge. It generally has a direct and beneficial impact on employment quantity through innovation, albeit this effect diminishes as businesses move towards the cutting edge of technology.

The Chinese experience is a bit different and process innovation exerts a positive effect on employment while product innovation has a negative effect. In particular, the rise in sales of outdated products has a favorable impact on job creation, whereas the rise in productivity of new products has a negative impact. Because diverse labor skill structures affect different enterprises differently, innovation has different effects (Chen et al., 2021). Some of the African experiences (Ghana, Kenya, and Zambia) have shared the conclusion somehow as evidenced by the study (Twumasi et al., 2022). The findings suggest that innovation boosts employment in the subregion. Furthermore, a more significant source of employment for businesses in sub-Saharan Africa is uniqueness in product invention.

Green innovation and sustainability

Knowledge has been progressively expanding in recent times at several stages, including limitation, accumulation, concentration, and diffusion. This has resulted in the formation of a distinct kind of invention, innovation, and entrepreneurship, which has allowed businesses to operate in extraordinary ways. The second wave focused on how knowledge, social and natural ecosystems, and state strategy could work together to offer the

theoretical framework for both practices and policies, current issues with society and development (Giorgos and Manolis, 2021).

Innovation is beyond growth perspective since it drives not only economic outcomes, but also addresses sustainable development which is linked to developments in the social, ecological, and economic spheres environmental and social sustainability. Green innovation is the application of innovative ideas to minimize harmful effects on the environment and increase sustainability, tackling global challenges such as climate change and resource depletion. Innovation for sustainable development is a recent phenomenon, but its development and implementation are equally complicated, dynamic, and uncertain as other forms of inventions. While it is admirable to want to maximize positive environmental consequences, such innovation must also be financially feasible (Bruno & Diana, 2019).

Technological accessibility, institutional frameworks created by governmental regulations to manage the transition, and a thorough analysis of the actors involved are all necessary for co-optimizing these advancements because innovation does not automatically translate into prosperity and a higher standard of living for all (Vollenbroek, 2001).

Given this background, He et al., (2023) and Harrison et al., (2014) assessed the effect of green technology innovation on employment from the viewpoints of environment sustainability and industrial structure optimization. The findings show that any form of innovative green technology can dramatically raise employment level. There also exists a positive relationship for firms introducing green innovation voluntarily, as compared to firms complying with regulations (Rasi and Ester, 2016). On the other hand, the green transition's effects on the labor market are probably going to be uneven across sectors and countries. There may also be significant local economic and distributional implications, which are linked to connected upstream and downstream industries. If labor market frictions prevent workers from reallocating their labor, the green transition may have a detrimental impact on overall employment on the other hand (Anneleen et al., 2022). Green innovation is like a two-edged sword when it comes to how it affects firm's productivity. Technological improvements and sustainability lead to a direct gain in productivity, while inefficiencies, adjustment costs, and disruptions—particularly in the short term—cause an indirect reduction in productivity (Hameed et al., 2024).

Method of the study

This study employed pseudo-panel data from the World Bank Enterprise Surveys Dataset for selected African countries (Ethiopia, Ghana, Cameroon, Rwanda, Zambia, Senegal, Cote d'Ivoire, Kenya, and Zimbabwe). It also has three methods of analysis to address the impact of innovation on employment and labor productivity.

Process of data management

In the data management process, some issues need to be focused in bold. These are:

- Taking all time dimensions: here, all-time dimensions are taken rather than focusing on the last two periods since there is no consistency in the panel data, par-

ticularly in innovation-related variables. For instance, one of the indicators of innovation, h1 (new product), has an observation for only 2006 and 2016 but not for 2011 in the Cameroon dataset, while it has an observation only for 2016 in the Cote d'Ivoire dataset. This problem exists in most innovation indicator variables, including new processes, new markets, and research and development; and it exacerbates employment-related variables, particularly those that categorize employment as production, non-production, skilled, or unskilled.

- Pseudo-panel based on legal status, firm size, and industry type: the exogenous variables to be used for the pseudo-panel, as scholars including Marine (2017) argue, should be common for all individual datasets in all countries. Hence, here it is found that only three exogenous variables are common for all single datasets in all countries. That is why the pseudo-panel is based on legal status, firm size, and industry type.
- The cohort is one, the average value for the continuous and other variables being one: the third issue in the data management process is the cohort. In the pseudo-panel estimation process, one cohort is generated using the common exogenous variables: legal status, firm size, and industry type. This helps us to get the group average value of continuous variables based on the cohort. For example, the pseudo-panel will give the group average value of full-time permanent employees for sole proprietorship, small, and food industries as one variable; the group average value of permanent employees for limited companies, large, and basic metals industries as another variable, and so on.
- Create a dummy variable for innovation: as mentioned above, since there is no consistency in the panel dataset, particularly in innovation and employment-related variables, which are the main interest variables in this study, a new dummy variable for innovation named innovation dummy is generated. If there is no innovation indicator variable, out of the nine innovation-related variables in each country's dataset, the value of innovation_dummy will be zero unless it is one. This dummy variable will be used to see the impact of innovation on labor productivity and employment using propensity score matching.
- Development of innovation as a counting variable: developing innovation as a counting variable in addition to generating a dummy variable for innovation is the other issue here. This will be used for the dose–response model estimation to see the impact of innovation on labor productivity and employment as the innovation dose increases.
- Computed productivity from employment and sales: one of the main variables of interest, labor productivity, is computed as the ratio of total annual sales and employment based on the recommendation from project leaders. The productivity of permanent full-time employees, productivity of production and non-production workers, and productivity of skilled and unskilled workers are also computed even if there are a large number of missing observations, especially concerning the productivity of skilled and unskilled workers.

- Develop one index for external bureaucracy variables (institutional obstacles or barriers) using factor analysis and polychoric correlation since the variables are categorical. In addition to innovation, it is expected that other factors may have a potential impact on labor productivity and employment. One of the factors can be external bureaucracy (institutional obstacles). However, since the number of variables under this category is large (more than 12), a single index for institutional obstacles is developed using exploratory factor analysis. In addition, since the variables under this category are ordered, polychoric correlation is used rather than Pearson correlation since Pearson correlation will be biased for non-continuous variables.
- Missing values: there are a large number of missing observations in almost all interesting variables. After removing missing values, the number of observations went from 11 thousand to 2923 based on the main variables. Based on firm idstd, we have 1972, but 1168 based on Panelid for innovation, productivity, and employment are cleared, without touching control variables. The descriptive statistics are based on 2923 observations after removing the missing values of the main interest variable, innovation, without touching other interest or control variables.

Method of data analysis

The study employs a quantitative approach, mainly using PSM to control for selection bias and associated endogeneity problems, and the dose–response model to analyze the impact of innovation intensity on labor outcomes. Data were derived from firms in several African countries, ensuring a broad representation of different industry types and sizes. The methodology is robust, providing a detailed examination of the nuanced effects of innovation across various labor segments.

Descriptive analysis

In this part, the study uses statistical summary for continuous variables; frequency distribution for categorical variables; correlation coefficient for examining the association of two continuous variables; Chi2 for capturing the dependency of two categorical variables; point biserial test for investigating the relationship between two continuous and categorical variables; and finally, it uses two sample tests to examine the impact of innovation on labor productivity, employment, and other important variables.

Propensity score matching method

As the two-sample *t*-test failed to figure out the specific impact of treatment on the outcome variables, the propensity score matching method is important in this regard to estimate the specific impact of interventions, exposures, and treatment on outcome variables. It identifies the treatment variables, outcome variables, and covariates (not affected by the program) and computes the probability of treatment on the condition of the observed covariates to balance the covariates between the treated group and the

control group, which are the statistical comparison groups. It also allows one to get a specific and unbiased estimate of the average treatment effect (Rosenbaum & Rubin, 1983). The computed probability helps to obtain similarities between treated and control groups and matches them based on observed characteristics. The average treatment effect of the intervention can be calculated after matching the treated and non-treated groups given covariates (confounding) that predict receiving the treatment in the specification of the model.

In this context, this study takes innovation as a treatment variable and labor productivity and employment as outcome variables, while some characteristics of firms are taken as covariates. The propensity score matching then reduces selection bias that may arise due to a difference in the treatment outcome between the two groups and may be triggered by factors that predict treatment instead of treatment itself. The randomized experiments of the PSM help to reduce treatment assignment bias (unbiased estimation of average treatment effect) as a result of matching the two groups comparable concerning the control variables. Thus, the PSM employs a quasi-experimental model and computes the average treatment effect (ATE) as it is impossible to compute the individual treatment effect.

According to Heckman et al. (1997), $ATE = E(Y) - E(Y)$ (this estimate might not be significant to policymakers for the reason that it includes the effect for whom the involvement was never intended). As a result, the most important valuation parameter is known as the average treatment effect on the treated (ATT) $= E(Y|D=1) - E(Y|D=0)$. This measures the effect on those for whom the intervention is intended. In a non-experimental study, one has to introduce some identifying assumptions to solve the selection problem. Once the two groups are matched based on observation, the average treatment effect of the intervention can then be calculated as the average difference in outcomes between the treated and non-treated units after matching across treated and control groups. This propensity value is estimated based on a statistical model, e.g., a logit or probit model, and thereby estimates the average treatment effect of the outcome difference between the two groups using nearest-neighbor, caliper, stratification, and kernel matching.

The other assumption is that there is conditional independence and common support. This overlap condition is defined as $P(T_i = 1|X_i) > 0$. This condition ensures that treatment observations have comparison observations “nearby” in the propensity score distribution. Specifically, the effectiveness of PSM also depends on having a large and roughly equal number of participant and nonparticipant observations so that a substantial region of common support can be found. For estimating the TOT, this assumption can be relaxed to $P(T_i = 1|X_i) > 0$.

Treatment units will therefore have to be similar to non-treatment units in terms of observed characteristics unaffected by participation; thus, some non-treatment units may have to be dropped to ensure comparability. However, sometimes a nonrandom subset of the treatment sample may have to be dropped if similar comparison units do not exist. This situation is more problematic because it creates a possible sampling bias in the treatment effect. Heckman et al. (1997) encourage dropping treatment observations

with weak common support. Only in the area of common support can inferences be made about causality. This reflects a scenario where common support is weak.

Different matching criteria can be used to assign participants to non-participants based on the propensity score. Specifically, nearest-neighbor matching, caliper or radius matching, stratification or interval matching, and local linear matching. One risk with this method is that only a small subset of nonparticipants will finally satisfy the criteria to fall within the common support and thus construct the counterfactual outcome. Non-parametric matching estimators such as kernel matching and LLM use a weighted average of all nonparticipants to construct the counterfactual match for each participant.

The quality of the matching procedure is evaluated based on its capability to balance the control and the treatment groups concerning the covariates used for the propensity score estimation. There are several procedures for this. The basic idea of all these approaches is to compare the distribution of these covariates in the two groups before and after matching the propensity score. If there are significant differences after matching, then matching on the propensity score was not successful in making the groups comparable, and remedial measures have to be taken.

A good matching procedure should reduce the standardized bias for each of the covariates used in the estimation of the propensity scores. Thus, this approach requires comparing the standardized bias for each covariate x before and after matching. A similar approach uses a two-sample t -test to check if there are significant differences in covariate means for both groups. After matching, the covariates should be balanced in both the treatment and the non-treatment groups. Therefore, no significant difference should be found in the distribution of the covariates between both groups. The pseudo- R^2 after matching should be fairly low. The same can be done by checking the F -statistics before and after matching. These statistics indicate the joint significance of all regressors used for the estimation of the propensity score (Caliendo & Kopeining, 2005).

It should be clear that matching estimators are not robust against these “hidden biases”. Different scholars have become increasingly aware that it is important to test the robustness of results to depart from the identifying assumption. Since it is not possible to estimate the magnitude of selection bias with non-experimental data, the problem can be addressed by sensitivity analysis. Sensitivity analysis method to address the issue of uncontrolled confounding in observational research. Propensity scores are obtained from observational data that lack randomization. The matching only controls for the differences in the observed variables, and there may be some bias resulting from the unobserved covariates that could affect whether subjects receive treatment or not. Estimation-based PSM is unbiased if there are no unmeasured confounders and if all relevant covariates have been incorporated into the estimation model. If there are unknown variables that affect both treatment assignment and the outcome variable, a hidden bias may occur for which matching estimators are not robust. The sensitivity parameter (γ) also measures the degree of departure from random assignment of treatment. It is a measure of departure from the study that is free of bias. In a randomized experiment, randomization of the treatment ensures that it is equal to 1, no sensitivity analyses are required. However, the value of γ is unknown for observational studies; sensitivity analysis requires a trial of several values of γ to see how the conclusion

might change. The concern in sensitivity analysis is whether or not an inference about treatment effect may be altered by unobserved factors.

Dose–response model

This model was proposed by Hirano and Imbens (2004) to evaluate the impact of programs or interventions with continuous treatment exposure. It is also an experimental condition and estimates a dose–response function using the generalized propensity score (GPS). It is an extension of the popular propensity score methodology for binary treatment group assignments (Rosenbaum & Rubin, 1983) and multivalued treatments (Imbens, 2000; Lechner, 2001). Indeed, Hirano and Imbens (2004) show that the GPS has a balancing property similar to the binary propensity score. Conditional on observable characteristics, the level of treatment can be considered random for units belonging to the same GPS strata. This means that adjusting for the GPS removes all biases associated with differences in the covariates.

A dose–response function (DRF) can be calculated using a regression approach when i) treatment is continuous; ii) individuals react differently to observable confounders; and iii) treatment selection is endogenous. From a program evaluation perspective, indeed, what is relevant in many settings is not only the binary treatment status, but also the level of exposure (or dose) provided by a public agency. The DRFs are usually estimated to check respondents' resilience to different levels of intervention.

Consider a policy program where the treatment is not assigned randomly (that is, it is assigned according to some structural rule) and where—after setting who is treated and who is not—the program provides a different level of exposure to treatment, ranging from 0 (no treatment) to 100 (maximum level of treatment). Two groups of units are thus formed: untreated, whose level of treatment (or dose) is 0, and treated, whose level of treatment is greater than 0. We are interested in estimating the causal effect of the treatment variable t on an outcome y within the observed sample, assuming that treated and untreated units may respond differently both to specific observable confounders (that we collect in a vector x) and to the intensity of the treatment t . We wish to estimate a DRF of y on t when the treatment is assumed to be either exogenous (that is, selection into treatment depends only on factors observable to the analyst) or endogenous (that is, selection into treatment depends on factors both observable and unobservable to the analyst).

Compared with similar models—and in particular, the one proposed by Hirano and Imbens (2004) and implemented in Stata by Bia and Mattei (2008)—this model does not need a full normality assumption, and it is well suited when many individuals have a treatment level of 0. Moreover, it may account for treatment endogeneity by exploiting instrumental variables (IV) estimation in a continuous treatment context. When many units are not exposed to treatment, the distribution of t has a “spike” or non-nil probability mass at 0, that is, $p(t=0) > 0$. As a result, assuming that the distribution of $t|x$ is normal (or a mixture of normal distributions), as assumed in Hirano and Imbens's (2004)

generalized propensity score (GPS), is untenable, because in the presence of a spike at 0, this distribution is discontinuous and thus nonnormal. Guardabascio and Ventura (2014), on the other hand, recently proposed a modification to the Hirano–Imbens model that extends the GPS approach to the case of a nonnormal continuous treatment variable. The authors consider a set of alternative distributions (binomial, Poisson, gamma, inverse Gaussian, etc.) derived from the exponential family of distributions. Similarly, Bia et al. (2014) have proposed a generalization of the Hirano–Imbens model that allows one to estimate semiparametrically the DRE. In their article, the authors propose to estimate the GPS parametrically under various alternative distributional assumptions (such as the normal, inverse Gaussian, and gamma distributions) using as link functions the identity function, the logarithm, and the power function. Moreover, in the case of a treatment variable assuming values in the (0, 1) interval, a two-parameter Beta distribution is also implemented.

Another interesting approach to modeling a continuous treatment setting is the one proposed by Adorno et al. (2007). They propose a nonparametric two-step matching approach based on the generalized propensity function developed by Imai and van Dyk (2004). In the first step, they specify a participation decision rule and match units on the set of covariates that identify such a process; in the second step, by considering only matched units from the first step, they perform another matching procedure, pairing units having similar values only for the covariates explaining the treatment level assignment. Treated and untreated units are thus assumed to be balanced in both processes (binary selection and level assignment) because the only difference between the two groups should be in program participation. Although this model considers the presence of “zeros” in the treatment level variable, it is suitable only under conditional independence.

However, the approach presented in this article has some of its limitations. First, differently from the semiparametric approach adopted by Adorno et al. (2007) and Bia et al. (2014), the approach presented here assumes a parametric form of the potential outcomes with additive separability. Second, it does not need an estimation of the GPS, which might be interesting to analyze itself. Third, the proposed IV estimation uses a Heckman bivariate selection model, which requires additional distributional assumptions. Fourth, it only considers the presence of observable heterogeneity, thus neglecting the possible presence of unobservable heterogeneity. The model is, however, well suited to address DRF estimation when conditions (i) and (ii) above are present.

Variable choice and definitions

Variable	Type and definition	Measurement
Outcome variables	Labor productivity	Total sales per employment (production, non-production, skilled and unskilled)
	Employment	Number of employment (production, non-production, skilled and unskilled)
Treatment variable	Innovation, a dummy variable	Innovation (overall, product, process, and RD) 1 = if any innovation and 0 = if none of them
	Innovation, dose	Innovation as a dose ranged from 0 to 10

Variable	Type and definition	Measurement
Possible covariate variables	largown_sha	What percentage of this firm does the largest owner(s) own?
	private_dom_a	% owned by private domestic individuals, companies or organizations
	private_fo~a	% owned by private foreign individuals, companies or organizations
	gov_sha	% owned by government/state
	oth_sha	% Owned by other
	mainpro_pa~e	Main product/service: percent of total annual sales
	psales_na	% of sales: national sales
	psales_inex	% of Sales: indirect exports
	psales_dex	% of sales: direct exports
	f1	In the Last F.Y., what was The Capacity Utilization (%) Of This Establishment?
	f2	Number of Hours Per Week Operated by The Establishment in Last Fiscal Year
	av_femp_est	mean_Number of Full-Time Employees of The Establishment When It Started
	av_powerot_m	mean_Number of Power Outages Experienced in A Typical Month in Last F. Y
	av_powerot_l	mean_How Long Did the Average Power Outage Last?
	av_elec_gen	mean_% Electricity from Generator Owned/Shared by The Establishment in Last F. Y
	av_totansa~s	mean_In last fiscal year; what was its total annual sales
	Firm size	Firm size
	legal status	Legal status
	quace_int	Quality certificate (international)
	power_short	Over the last F.Y., Did This Establishment Experience Power Outages?
	generator	Was Generator Shared Or Owned Over The Last Fiscal Year?
	own_web	Own website
	tec_foreign	Technology: foreign licensed
	training_pfemp	Formal training: full permanent employees
	extbrcy	External bureaucracy (institutional obstacles)

Results and discussion

Descriptive statistics (refer Appendix A)

In this section, the study presents summary statistics, frequency distribution, Pearson correlation coefficient, Chi2 tests, point biserial tests, and two-sample *t*-tests. These all help to understand the behavior and distribution of each variable, their association and dependence, and the simple impact of innovation on labor productivity, employment, and other most important variables.

Except for Zambia and Cote d'Ivoire, on average, more than 80% of the sample firms belong to private domestic owners. The share of foreign private shareholders is below 9% on average for Ethiopia, Cameroon, Kenya, and Zimbabwe, while it is more than 20% for Zambia and Cote d'Ivoire. On the other hand, it is not only the share of foreign shareholders but also the government share that is less than 1% on average in all countries. This may suggest that the involvement of foreign individuals, companies, or

organizations in developing countries is not as widespread as it may have been, and this may have a potential impact on labor productivity and employment. This may be because foreign individuals, companies, and organizations may have more access to modern methods of products, technologies, and organizational developments, which may affect the productivity of employees directly or indirectly.

This low percentage of shares for foreign individuals, companies, or organizations in all sample countries may also be related to the ease of doing business in these countries. This is because unless the institutional obstacles, including the bureaucracy in export and import licenses, access to telecommunication, electricity, water, finance, land and transportation, customs and trade regulations, and security are improved, foreign investors will be reluctant to come and invest in these developing countries. On the other hand, unless governments are committed to improving the quality of institutions and bringing incentives for foreign investors, the participation of foreign companies and organizations in domestic investment will be very low since those foreign investors will go somewhere else, which enables them to get a higher return from their investment. This will affect not only innovation and hence labor productivity, but it may also affect the overall economy through a shortage of hard currency.

On average, in almost all countries, the number of hours per week operated by firms is more than 56 h. The highest number of hours per week operated by the firms is in Cote d'Ivoire, with 71 h per week. This implies that in almost all countries, firms are operating in less than two shifts and less than eight hours per day. This may also have an impact on productivity, since the fewer the hours operated by firms, the lower the output will be. In addition, employees will spend their time fulfilling the day-to-day ration of products and services rather than giving time for new ideas and developments. Hence, the probability of innovating new products or services, new marketing strategies, or new organizational structures will be low. Except for Kenya, where the average percentage of sales through direct and indirect exports exceeds 10%, more than 90% of sales in other countries are conducted in domestic markets. This may indicate that the export intensity, which is measured as the ratio of aggregated export sales to total firm sales, is very low in all sample countries. This can be related to the type and quality of products and services, which influence the competition in the international market.

The food, wholesale, retail, and logistic services industries, particularly hotels and restaurants, account for half of the total sample industry types from which the data were collected. On the other hand, only a small number of firms from four countries were selected for the data collection in the tobacco and refined petroleum industry types. In other words, among the sample countries, only three countries (Cameroon, Kenya, and Zimbabwe) have firms working on refined petroleum, and two countries (Ethiopia and Zimbabwe) have one firm each operating on tobacco. Vis-à-vis the total number of industries considered in the data collection, 60% of industries come from Kenya, Cameroon, Zambia, and Ghana, while only 9% of the industries come from Rwanda and Cote d'Ivoire.

The firm's size by country and year out of the total firms, only 2% are micro-enterprises, while 48% are small. Furthermore, 60% of firms in Ghana, Cameroon, Zambia, and Senegal are small businesses. Ethiopia, Kenya, and Zimbabwe are countries in which more large industries are considered compared with other countries. Table 6 presents

the legal status of industries by country. According to this table, sole proprietorships or limited companies account for 68% of the industries.

Only 16%, 39%, and 37% have international quality certificates, their websites, and prepare and provide formal training for their permanent full-time employees, respectively. Even if it is proved in the inferential analysis, in this modern world in which the market is full of competition, international quality certification, having a website and formal training for employees may have a potential impact on labor productivity, market share, and sales, and the performance of firms in general. Compared with other countries, Senegal comes last with the lowest achievement, having only 7% and 16% of firms with international quality certificates and training their employees, respectively, and Cameroon, with only 20% of its sample firms developing their website, which is important for communication, marketing promotion, and development.

In terms of innovation, as shown in Appendix A, more than 90% of sample firms record at least one of the nine types of innovation, which include new products or services, new marketing strategies, new production processes, and new organizational structures. Here, except in Cameroon, where around 49% of the sample firms do not record any of the innovation types, all sample firms in all countries are working on innovation. The other issue here is that more than 54% of sample firms use a generator for their operations since more than 91% of them report a power shortage. This makes the production costs high and hence decreases the performance of those firms.

The number of innovation types recorded in each sample country is shown in Appendix A. Only in one country, Cameroon, do 60% of firms lack product innovation, as shown in the table. However, more than 89% of the total sample firms have a product innovation. This means that this percentage of businesses create new products, new services, or new ways of manufacturing or providing services. In terms of process innovation, 70% of the total sample firms innovate either new logistic, delivery, or distribution methods; new or improved management practices; new marketing methods; or new organizational structure. In this type of innovation, the highest percentage (more than 95%) of firms in Rwanda innovate new methods of logistics, marketing management practices, or organizational structure, while the lowest percentage (21%) is recorded in Cameroon. Regarding research and development, all sample firms in Zimbabwe are working on it. 55% of the total sample firms invest time and money in research and development to create new products, services, or methods.

The correlation between continuous variables shows that there is a negative correlation between labor productivity and the percentage of private domestic share, the percentage of government share, and power shortage, while the correlation is positive with the percentage of private foreign share. This may imply that (need to prove in the inferential analysis) the larger the share of the private foreign shareholders enables them to use modern technologies, arrange training for their employees, and hence increase productivity. However, the higher the share of the government may decrease productivity, particularly when there are several institutional obstacles, which are obvious in Africa.

The correlation between innovation and other categorical variables, including firm size, legal status, owned website, and international quality certificate, is strong. As a result, it is possible to say that a relationship exists between innovation and firm size, innovation and legal status, innovation and own website, innovation and international

quality certificate, innovation and formal training for employees, and innovation and foreign licensed technology. When employees get capacity building through training, they may be encouraged to try new ideas and hence innovation. In addition, the larger the firm is, the higher the capacity to spend on research and development, which may encourage employees to spend their time on innovation.

Using the point-biserial correlation test, Appendix A reveals the correlation between categorical and continuous variables, mainly between innovation and labor productivity, employment, and other continuous control variables. As the result shows, there is a significant positive relationship between innovation and the percentage of private foreign shares and full-time permanent employees. This implies that the larger the share of private foreign individuals, companies, or organizations, the higher the probability of innovation since foreign shareholders have access to modern technologies, knowledge, and new ideas compared to domestic private shareholders, and hence, they can arrange training for their employees and encourage their employees to spend more time on innovating new products, new marketing strategies, and research and development. In addition, permanent employees have more access to training and can spend more time on research and development (if the working environment allows them) compared to seasonal or temporary employees. Hence, permanent full-time employees may have a significant positive impact on innovation.

On the other hand, innovation has a significant negative relationship with the percentage of other shares (shares not owned privately or by the government), the frequency of power shortages per month, and external bureaucracy (institutional obstacles).

The impact of innovation on employment and labor productivity using a simple *t*-test is shown in Appendix A. It is revealed that innovation has a significant impact on employment, main product/service as a percentage of total annual sales, and external bureaucracy. However, the two-sample *t*-test does not consider whether the impact is due to and specific to the treatment variable. Therefore, the propensity score matching and then dose–response models are recommended to do so.

Results of the propensity score matching

Impact of the overall combined innovation (refer Appendix B)

This section presents, using propensity score matching (PSM), the overall combined effects of all types of innovations (product, process, and R&D) on the employment and productivity of various workforce segments. The findings indicate that while all forms of innovations—process, product, and R&D—have a favorable overall impact on the productivity of production workers, they hurt the productivity of permanent, non-production, skilled, and unskilled workers. This PSM result bolsters the claim made by Mohnen and Hall (2013), Duguet (2006), and Roper et al. (2008) that innovation may lower labor productivity in its early phases because of the new product's deriving effect on the old one and the disruption effect at the beginning caused by inefficient production. Furthermore, this result is consistent with the descriptive analysis, which indicates that labor productivity may be significantly harmed by innovation. However, as Appendix B illustrates, innovation may have a major positive effect on production workers' productivity. While innovation contributed to increased employment, particularly in the production sector, it led to a decrease in labor productivity, especially among skilled and

non-production workers. This suggests that the benefits of innovation in terms of job creation are offset by initial inefficiencies or disruption in work processes.

This result supports the claim made by Mairesse, and Sassenou (1991), Hall et al. (2009), Mairesse and Mohnen (2010), and Mairesse et al. (2005) that innovation, especially when it involves a new product or service, can significantly increase labor productivity by meeting market demand, lowering unit costs, and utilizing economies of scale. Several academics, such as Avenyo et al. (2019), contend that innovation may have a larger effect on job growth than labor productivity alone. The PSM results showed that innovation has a statistically significant overall and combined effect that benefits skilled and non-production workers, suggesting that innovation increases demand for these workers. This result is also in line with the research by Guillermo et al. (2020), which found that high-skilled workers benefit significantly more from innovation in the workplace than do low-skilled people. The findings imply that innovation, particularly in the area of products, promotes a bias in favor of high-skilled job demands at the expense of low-skilled individuals. It is also consistent with the claims made by Avenyo et al. (2019) that innovation has the potential to greatly boost employment, especially in medium- and low-tech industries where sub-Saharan Africa is well-known for its firm structure. According to Okumu et al. (2019) and Cirera and Sabetti (2016), additional sales linked to innovation are probably going to be produced at the same or higher labor intensity. Furthermore, in sub-Saharan Africa, where businesses are less connected to cutting-edge technology, this noteworthy beneficial impact is far stronger (Cirera & Sabetti, 2016).

Impact of the product innovation (refer Appendix C)

The productivity of all labor segments—permanent, production, non-production, skilled, and unskilled workers—is statistically significantly impacted by product innovation, but only the productivity of production workers is positively impacted. This raises some concerns about why it negatively affects other segments of labor. Product innovation may potentially have a short-term negative impact on productivity since changes take time to implement. Furthermore, it might take a while for new items to get market recognition and there is no guarantee that they will be successful on the market. Additionally, it has been proposed that the detrimental effects of tailored production on productivity are likely to be triggered by new products (Roper et al., 2008; Tomasz & Arkadiusz, 2018).

Furthermore, except for skilled workers, product innovation has no statistical significance and little ability to affect any segments of the labor market. This can be reasoned out to the fact that the net effect of direct effect and indirect can be fully settled by each other in quantitative terms. This means that new items that are manufactured more efficiently than old products may require less labor input for a given output. This can nullify the impact of product innovation that has a positive direct labor effect by creating new demand (Pianta, 2005; Vivarelli, 1995). Furthermore, the impact of new products on labor demand varies depending on whether they are complementary or substitutes. For example, a firm needs less labor if the demand for new items can partially replace the demand for old products. If they are complementary, the firm needs more work

otherwise. Therefore, if firms in industry engage in both sorts of products, the effect could be null on employment on average (Peters et al., 2017).

Impact of the process innovation (refer Appendix D)

It shows that, on average, process innovation has a statistically significant positive impact on the productivity of production workers, while hurting the productivity of skilled, non-production, and permanent workers. After matching, the productivity of the skilled, unskilled, and nonproduction-treated groups is lower than that of the non-treated groups. The sole distinction is that the impact's magnitudes vary depending on the kind of labor group.

This PSM conclusion further reinforces the claim made by Hall (2011) and Roper et al. (2008) that early innovation may lower labor productivity because of the way the new depends on the old and because of the disruption caused by inefficient manufacturing. Furthermore, this result is consistent with the descriptive analysis (see Table 3b in Appendix A), which indicates that labor productivity may be significantly harmed by innovation.

Conversely, innovation could significantly increase the productivity of workers in the production area. After matching, it is discovered that the treated groups are more productive than the non-treated groups. This result supports the claims made by Mohnen and Hall (2013) and Masso and Vahter (2008) that process innovation, which is positively connected with lower transaction costs, may boost worker productivity. Furthermore, research from Hall, Mairesse, and Sassenou (1991), Hall et al. (2009), Mairesse and Mohnen (2010), and Mairesse et al. (2005) suggest that innovation, especially when it addresses a new market demand, lowers unit costs, and takes advantage of economies of scale, may have a significant positive impact on labor productivity. All of the aforementioned data collectively demonstrate the fallacy of claims made regarding the relationship between labor productivity and innovation.

Process innovation has a positive effect on the employment of skilled, non-skilled, and production workers. This result is consistent with the claims made by Avenyo et al. (2019) that innovation has the potential to greatly boost employment, especially in medium- and low-tech industries where sub-Saharan Africa is well-known for its firm structure. The claims made by Okumu et al. (2019) and Cirera and Sabetti (2016) all support the idea that process innovation has the potential to increase employment because the new sales it generates will probably have a labor intensity that is equal to or higher than before.

Impact of R&D innovation (refer Appendix E)

Permanent, non-production, skilled, and unskilled worker productivity does not positively react to R&D innovation. Given that internal and external circumstances may cause its impact to occur early on, this negative influence may materialize in the short run. Given that long-term economic growth is thought to originate from R&D innovation, it might end up being advantageous in the long run. This result goes against the claims made by Mairesse, and Sassenou (1991) and Mairesse and Mohnen (2010) and is in support with the claims made by Hall (2011) and Roper et al. (2008).

Research and development is the creative expand the body of knowledge and come up with new uses for the knowledge that is already available so it takes time to see its beneficial effects on performance by creating a significant competitive advantage. In the long run, higher R&D expenditure results in higher production efficiency. It is critical to recognize that research and development is only one source of innovation. Not all breakthroughs come from R&D and not all science, research, and technology advance productivity. Innovations increase productivity by reducing manufacturing costs or raising the caliber of products and services. Certain innovations, such as those that result in higher-quality products and services, could not have a significant impact on increasing productivity as measured. Still, some of these—like technology advancements in elder care—lead to better health and quality of life. Sometimes, gaining more knowledge can have significant and unanticipated effects on productivity.

This finding defies the previous one in that R&D significantly increases the employment of production, non-production, skilled, and unskilled people. R&D may greatly increase employment, especially in sub-Saharan Africa, where medium- and low-tech industries predominate, according to Cirera and Sabetti (2016), Okumu et al. (2019), and Smolny (1998). This is because new sales associated with R&D innovation are likely to be produced with the same or higher labor intensity. The case made by Avenyo et al. (2019) further supports the idea that innovation may have a positive effect on the expansion of employment in Africa.

Results of the dose–response estimation (refer Appendix F)

This study uses a dose–response estimation technique in addition to PSM to examine the possible effects of treatment intensity on labor productivity and employment in the chosen sub-Saharan African nations. According to the plot of the dose–response function (DRF), the relationship between intensity and productivity of permanent employees is strongly declining and fairly accurately estimated for lower values of the dose, for higher levels of the dose, the relationship is more strongly declining but less accurately estimated. Considering the remaining DRFs, the productivity of production, non-production, skilled, and unskilled workers decreases as innovation intensity increases. All of these findings are in line with the propensity score matching method (PSM), except for how combined and total innovation affects production workers. In other words, the intensity has a declining DRF effect while innovation has growing (positive) effects by PSM for production workers. Some of the findings are in favor of Mohnen and Hall (2013), Duguet (2006), and Roper et al. (2008) arguments that, particularly at the initial stages of innovation, labor productivity may diminish in innovative firms. In addition, most of the findings are consistent with the above arguments under PSM and descriptive statistics, which attests to the robustness of the estimation results.

Aside from innovation (the treatment variable), firm age, having an international quality certificate, total main product, foreign private ownership, and number of working hours per week all have a significant positive impact on labor productivity. Nonetheless, power shortages, government ownership, and domestic private ownership decrease the productivity of employees significantly. This can be because foreign individuals, companies, and organizations may have more access to modern methods of products, technologies, and organizational developments, which have a

potentially positive effect on labor productivity compared to domestic private companies, organizations, and governments. This argument is consistent with the argument from descriptive statistics that foreign private ownership has a positive correlation with labor productivity. This finding is also compatible with the arguments from Griffith et al. (2004), and Mebratie and Bedi (2013) that firms' age, foreign ownership, and internationally recognized quality certificates have a significant positive impact on labor productivity.

Regarding intensity impact on employment, the DRF also indicates that there is a strong relationship between the intensity of innovation and permanent employees as dose increase with quite precisely estimation on average. On the same trend, the impact of intensity has also increasing effect on the employment of skilled workers. However, it has a decreasing effect at the initial stage of the dose, but an increasing effect as the dose increases in the middle, and then starts to decrease around the full dose in the case of production workers. But the general trend is increasing. For both non-production workers and unskilled workers, the impact decreases at the initial stage but then increases. However, the intensity of innovation drives increasing effects for skilled workers. In general, the dose–response analysis showed that higher innovation intensity amplifies the positive employment effects, especially in firms engaging heavily in innovation. However, increased innovation also correlates with a greater negative impact on labor productivity, particularly in the early stages of adopting new processes or technologies.

Some of these findings are in line with the arguments from Avenyo et al., (2019), Cirera and Sabetti (2016), Okumu et al., (2019), and Smolny (1998) that innovation may have a significant positive impact on employment growth. This will happen when the new sales associated with the product/process/market/service innovation come with higher labor intensity. This significant positive impact is much higher in countries where firms are far from the technological edge. This argument is also consistent with the findings from propensity score matching (PSM). However, the employment of unskilled production workers decreases as the dose of innovation increases. However, Fig. 1:i (ref. Appendix E) shows that at first there is a negative relationship between innovation and employment of unskilled production workers, but then after around five doses of innovation, their relationship becomes positive. This also shows the inconsistency of arguments on employment and innovation.

Besides innovation, small, medium, and large firms have a significant positive impact on employment compared with micro firms. Having their website, the age of the firms, having an international quality certificate, having a generator as a substitution for lack of electricity, government ownership, and foreign private ownership has also had a significant positive impact on employment of full-time permanent employees and employment of production and non-production workers. The arguments from Adelino et al., (2017) and (Jabir & Nadia, 2021) also attest that existing firms will have a significant contribution to job creation through their expansion. Firms with an internationally recognized quality certificate create more jobs compared with those firms without the internationally recognized quality certificate, and employment growth is relatively high in foreign-owned establishments. However, the employment of skilled production workers decreases as the total sales increase through indirect export and national sales as a percentage of the total sales.

Conclusion and the way forward

African countries have been attempting to face the deeply rooted problems of perpetual economic growth and economic transformation through a range of reforms and redesigned industrialization policies to enhance firms' productivity, widen employment opportunities, and establish a broad-based economy. To do so, innovation is considered one of the integral parts and the means to sustain the growth of manufacturing sectors. It consists of various innovations, including product innovation, process innovation, and RD innovation, by converting ideas and knowledge into production processes through learning by doing, experiences, exposures, attitudes, skills, and economies of scale, which lead to enhanced market position and economic growth in the framework of a knowledge-based economy.

The study employs World Bank Enterprise Survey data for Ethiopia, Ghana, Cameroon, Rwanda, Zambia, Senegal, Cote d'Ivoire, Kenya, and Zimbabwe and creates pseudo-panel data with the specific objective of examining the impact of innovation on employment and labor productivity using propensity score matching and the dose-response method. More than 90% of sample firms report at least one innovation, including a new product or service, a new marketing strategy, a new manufacturing process, or a new organizational structure. This means that 2,673 firms are engaged in at least one of the innovation activities and 250 firms do not have any innovation activities.

The study finds that process innovation has a statistically significant impact on production, non-production, and skilled employees. And, it has a positive effect on the productivity of production workers and a negative effect on the productivity of non-production and skilled employees. However, it has a positive and statistically significant impact on these three labor types in terms of employment. R&D innovation, on the other hand, has a 10% negative and statistically significant impact on the productivity of permanent, non-production, skilled, and unskilled workers. And, R&D innovation has a positive and statistically significant impact on creating employment opportunities for production, non-production, skilled and unskilled employees.

Finally, this research ends and adds a great deal to our knowledge of how innovation affects employment and labor productivity in African economies. It casts doubt on the notion that innovation always boosts output by exposing a more nuanced dynamic in which the creation of jobs is accompanied by decreased productivity in some labor markets. The results highlight how crucial it is to back up innovative policies that solve productivity issues and increase employment. For academics, decision-makers, and business executives who want to support innovation-driven growth in developing nations, this report is an invaluable resource.

Policy implications

Approximately 90% of the African firms that were chosen for the World Bank Enterprise Survey have been trying to continue engaging in innovative activities. They are, however, falling behind others in terms of depth of innovation. There is also remarkable progress in certain activities though it results in a mixed effect on productivity and job creation.

On its own, innovation is beneficial and has the potential to have a significant positive economic impact. However, in contrast to its potential, this study finds contradictory results regarding the influence of innovation on employment and productivity in the African environment. One of the policy implications is that there must be a strategy of capitalizing on the positive impact of innovation on the productivity and employment of different labor segments, and think about the way how to reverse the negative impact of innovation on productivity and employment by providing due attention to the driving factors and explanations behind the results. It means that it should discover the factors and explanations that might diverge or converge with their possible influence on employment and productivity such as lengthy adjustment time, market imperfection, returns of innovation, and complementarity and substitutability nature of innovation products. Consequently, the results, as well as these explanations, must be considered when designing innovation policies, labor market policies, and economic development initiatives in the chosen nations. Routines are a firm's genetic code, yet innovation policies and labor strategies are not as detail-oriented in Africa. Therefore, rather than implementing uniform and aggregated policies with a linear approach, the policy should capture the complex character of innovation and its accompanying impact on productivity and labor forces. So, the study would inform the policy settlements to be disaggregated into distinct segments of labor outcomes.

Rethinking how innovation is adjusted and how it deviates from its intended purpose in increasing productivity and job creation should also be part of the policy prescription. This problem deals with shortening the time it takes for a product to go from idea to market availability, which promotes a positive indirect effect of innovation. The adjustment period is too protracted in the African context, resulting in a long lag structure. This would result in a high transaction cost, defects, wastes, and detrimental effects from both internal and external business-facilitating factors in the activities of product and R&D innovation. To counteract the detrimental effects of specific product and R&D innovations, policies and strategies should take this into account and shorten the adjustment time and lag structure by expanding provisions like soft and hard infrastructures. Otherwise, it leaves the market success of new products with uncertainty.

Besides, there must be a policy framework to manage both the direct and indirect effects of innovation to reverse its negative influence on productivity and employment. In actuality, the majority of African policies prioritize the direct impact above the indirect one. Product innovation is supposed to have a positive direct labor effect, whereas process innovation is thought to have a negative influence on employment. However, the indirect effects (i.e., the response or feedback of the market to innovation itself) might mitigate the favorable and unfavorable labor outcomes. For instance, the increased labor demand for product innovation might be challenged by the indirect effect if the new products are more efficient, and reduce the labor demand. In the same fashion, the negative direct labor effect of process innovations can be reversed by the indirect effect of

increasing production efficiency. In short, the response of the market and its capability of transmission to influence productivity and employment must be considered in the policy framework. The key question is: in what market conditions is innovation being implemented? Since wage and price rigidity is the primary element influencing innovation's function, the policy framework should take this into account depending on the specific nature country's context. Furthermore, the type of a good—whether it is substitutable or complimentary to an existing product—has implications for how innovation affects employment. While the complimentary character increases the demand for labor, the substitutability nature reduces the requirement for more labor. So, the policy framework should reconsider this issue and design a compensation mechanism accordingly to keep productivity and employment vibrant.

The nature of the information obtained from innovation surveys is the subject of part of policy implication. There are many reasons to reconsider it. Among them are, but are not limited to, the dummy variable's intrinsic imprecision for innovation, the absence of pertinent variables, and productivity measurements have limited doing a thorough study that accurately reflects reality. Therefore, to obtain relevant data and wise policy recommendations.

(see Tables 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29, 30, 31, 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48).

Limitations and areas of future research

One of the study's shortcomings is that the relevant data for the African nations that were left out were unavailable. The World Bank Survey's content also prevents the computation of alternative measures for a firm's productivity and innovation. In light of this restriction, future studies could concentrate on creating analytical techniques that integrate the general structural equation model (GSEM) and propensity score matching (PSM) to examine the mediating effects of adjustment time, market imperfection, and the complementarity and substitutability of products in the nexus of innovation, productivity, and employment. And, the future will be more demanding for having a comparative analysis of PSM and GSEM for those firms with and without green innovation.

Appendix A

Table 1 Innovation by country

Country	Gha	Eth	Cam	Rwa	Zamb	Seng	Cot	Ken	Zimb	Total
Innovation										
Yes	357	313	261	148	369	280	122	510	313	2673
No	0	0	250	0		0	0	0	0	250
Total	357	313	511	148	369	280	122	510	313	2923

Table 2 Types of innovation by country

Country	Gha	Eth	Cam	Rwa	Zamb	Seng	Cot	Ken	Zimb	Total
Product innovation										
Yes	357	313	204	148	369	280	122	510	313	2616
No	0	0	307	0		0	0	0	0	307
Total	357	313	511	148	369	280	122	510	313	2923
Process innovation										
Yes	319	213	110	141	294	236	42	479	213	2047
No	38	100	401	7	75	44	80	31	100	876
Total	357	313	511	148	369	280	122	510	313	2923
R&D innovation										
Yes	199	48	65	125	203	275	20	206	313	1454
No	158	265	446	23	166	5	102	304	0	1469
Total	357	313	511	148	369	280	122	510	313	2923

Table 3 Sum_ by country for continuous variables

Variable	Ghana			Ethiopia			Cameroon			Rwanda		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
av_pfemp_p	42,28866	1	540	142,9038	1	5000	54,9881	1	1500			
av_pfemp_np	10,56915	0	200	53,26282	0	717	16,75833	0	500			
av_pfemp_sp	27,48454	0	520	98,8129	0	1540	32,32534	0	709			
av_pfemp_unp	16,22034	0	300	15,99342	0	330	42,43713	0	927			
av_pfemp_npf	2.6	0	45	17,47134	0	300	3,646067	0	50			
ppw	151,3268	300	1,12e+07	1,317,940	3586.4	7,58e+07	1,89e+07	71,428.57	8,37e+08	3,14e+07	52,336.6	1,07e+09
pprw	109,7424	-47,777.7	4,000,000	1,423,120	13,440	9,18e+07	3,22e+07	153,846.2	1,60e+09			
pnprw	433,8388	1333,333	2,08e+07	1,953,944	26,666.6	2,65e+07	1,97e+08	250,000	4,50e+09			
psprw	314,894.7	-47,777.7	1,27e+07	1,616,346	-177,777	9,18e+07	1,27e+08	500,000	1,77e+09			
punsprw	297,134.5	500	1,00e+07	8,439,576	93,220.3	1,40e+08	6,48e+07	140,000	6,12e+08			
Variable	Zambia			Senegal			Cote d'Ivoire			Kenya		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
av_pfemp_p	33,19898	1	800	39,32039	1	770	127,7188	3	800	105,0466	1	7000
av_pfemp_np	14,32275	0	320	14,57426	0	128	46,41935	0	430	30,68539	0	1000
av_pfemp_sp	20,58673	0	500	23,6699	0	770				58,57706	0	3000
av_pfemp_unp	12,69792	0	300	14,64583	0	361	52,74074	0	500	49,98016	0	4000
av_pfemp_npf	3,27027	0	100	3,608696	0	85	6,766667	0	70	9,019231	0	400
ppw	1,69e+08	166,666.7	3,20e+09	1,51e+08	200,000	2,55e+10	9,21e+07	80,000	4,28e+09	1,28e+07	10,000	8,33e+08
pprw	2,32e+08	250,000	5,00e+09	3,53e+07	-	1,21e+09	9,73e+07	333,333.3	1,10e+09	2,65e+07	-3,11e+08	1,14e+09
					1,44e+09							
pnprw	7,78e+08	500,000	2,50e+10	1,02e+08	1,300,000	9,33e+08	2,42e+08	5,666,667	1,01e+09	7,33e+07	50,000	5,00e+09
psprw	4,29e+08	-	1,25e+10	7,55e+07	-	1,89e+09				6,81e+07	-3,11e+08	4,67e+09
		1,33e+07			1,44e+09							

Table 3 (continued)

Variable	Zambia			Senegal			Cote d'Ivoire			Kenya			
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	
punsprw	6.97e+08	250,000	1.20e+10	2.42e+08	400,000	3.38e+09	3.46e+08	1,800,000	1.80e+09	6.25e+07	16,666.67	4.00e+09	
Zimbabwe													
Variable	Mean						Min						Max
av_pfemp_p	142.9038						1						5000
av_pfemp_np	53.26282						0						717
av_pfemp_sp	98.8129						0						1540
av_pfemp_unp	15.99342						0						330
av_pfemp_npf	17.47134						0						300
ppw	1,317,940						3586.4						7.58e+07
pbrw	1,423,120						13,440						9.18e+07
pnprw	1,953,944						26,666.6						2.65e+07
							7						
psprw	1,616,346						-						9.18e+07
							177,777						
							8						
punsprw	8,439,576						93,220.3						1.40e+08

Table 4 Sum_ by year for continuous variables

Variable	2006			2011			2013			2014		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
av_pfemp_p	18.17731	0	500				65.79821	1	7000	39.32039	1	770
av_pfemp_np	32.32534	0	709				20.01087	0	1000	14.57426	0	128
av_pfemp_sp	60.22302	0	927				38.43049	0	3000	23.6699	0	770
av_pfemp_unp	4.813725	0	44				28.83092	0	4000	14.64583	0	361
av_pfemp_npf	1.79e+07	140,000	2.72e+08	3.14e+07	52,336	1.07e+09	5.515447	0	400	3.608696	0	85
ppw	2.40e+08	5,000,000	2.42e+09				8.08e+07	300	3.20e+09	1.51e+08	200,000	2.55e+10
pprw	1.27e+08	500,000	1.77e+09					– 3.11e+08	5.00e+09	3.53e+07	–	1.21e+09
pnpw	6.77e+07	140,000	6.12e+08				2.56e+08	1333.33	2.50e+10	1.02e+08	1,300,000	9.33e+08
psprw							1.57e+08	– 3.11e+08	1.25e+10	7.55e+07	–	1.89e+09
punsprw							2.48e+08	500	1.20e+10	2.42e+08	400,000	3.38e+09
Variable	2015			2016			2016			2016		
	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max	Mean	Min	Max
av_pfemp_p	142.9038	1	5000	75.05172	1	1500						
av_pfemp_np	53.26282	0	717	23.45133	0	430						
av_pfemp_sp	97.43949	0	1540									
av_pfemp_unp	15.99342	0	330	23.84375	0	500						
av_pfemp_npf	17.47134	0	300	3.40566	0	70						
ppw	1,317,940	3586.4	7.58e+07	3.56e+07	71,428.5	4.28e+09						
pprw	1,423,120	13,440	9.18e+07	4.76e+07	153,846	1.60e+09						
pnpw	1,953,944	26,666.67	2.65e+07	1.54e+08	250,000	4.50e+09						
psprw	1,616,346	– 1,777,778	9.18e+07									
punsprw	8,439,576	93,220.34	1.40e+08	1.51e+08	1,000,000	1.80e+09						

Table 5 Correlation between innovation and employment as well as labor productivity

Variable	Coefficient	P-value
Number of full-time permanent employees	0.0399	0.0321
Number of full-time permanent productive employees	0.0313	0.2788
Number of full time permanent non-productive employees	0.0442	0.1129
Number of full-time permanent skilled employees	0.0269	0.3517
Number of full-time permanent unskilled employees	− 0.0060	0.8325
Productivity of permanent employees	0.0197	0.3189
Productivity of full-time permanent productive employees		
Productivity of full-time permanent non-productive employees		
Productivity of full-time permanent skilled employees		
Productivity of full-time permanent unskilled employees		

Table 6 Impact innovation on labor productivity and employment using t-test

	Mean difference	P-value
Productivity of permanent employees	− 3.57e + 07	0.1594
Total annual sales: lfy	− 2.26e + 09	0.1175
Number of full-time permanent employees	− 38.91366	0.0160
Main product/service: percent of total annual sales	2.100922	0.0852
Number of full-time permanent productive employees	− 55.08599	0.1393
Number of full time permanent non-productive employees	− 12.36349	0.0564
Number of full-time permanent skilled employees	− 20.9749	0.1757
External bureaucracy	0.3533792	0.0168

Appendix B

The impact of overall level of innovation

Table 7 Impact of overall innovation on the productivity of labor

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	14.3962771	14.802733	− 0.4064558	0.2262974	− 1.80	Statistically significant at 10%		
	ATT	14.3962771	14.7891549	− 0.3928779	0.2154668	− 1.82	Statistically significant at 10%		
Sample logppwUnmatched		Ps R2 0.181	LR Chi2 167.54	P > Chi2 0.000	Mean bias 25.6	Med bias 17.3	B 115.0	R 0.41	%Var 100
Matched		0.010	29.41	0.001	4.0	3.3	23.3	1.56	100

Table 8 The impact of overall innovation on the productivity of production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	63,915,491.4	5,662,160.12	58,253,331.3	51,043,590.5	1.14	Statistically insignificant		
	ATT	63,915,491.4	6,208,981.66	57,706,509.7	13,730,242.8	4.20	Statistically significant at 5%		
Sample pprwUnmatched		Ps R2 0.212	LR Chi2 57.59	P > Chi2 0.000	Mean bias 29.7	Med bias 19.0	B 124.1	R 0.22	%Var 50
Matched		0.003	3.98	0.782	3.9	4.5	12.4	0.88	50

Table 9 The impact of overall innovation on the productivity of non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	15.860349	17.0969406	− 0.251248	0.3640016	− 3.44	Statistically significant at 5%		
	ATT	15.860349	17.5751633	− 1.729471	0.4247155	− 4.05	Statistically significant at 5%		
Sample logpnprw		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.164	69.31	0.000	31.5	20.9	104.1	0.37	100
Matched		0.001	1.49	0.960	2.8	2.7	7.5	1.21	25

Table 10 The impact of overall innovation on productivity of skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		15.2639389	17.3014418	− 2.037502	0.52016320	− 3.92	Statistically significant at 5%		
ATT		15.2639359	17.8167566	− 2.552817	0.54746767	− 4.66	Statistically significant at 5%		
logpsprw	Sample	Ps R2	LR Chi2	P > Chi2	Mean bias	Med	B	R	%Var
Unmatched		0.207	56.48	0.000	47.0	Bias 39.2	121.2	0.34	100
Matched		0.004	4.78	0.311	5.2	4.3	14.1	1.63	0

Table 11 The impact of overall innovation on productivity of unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	16.0532814	16.887689	− 0.8344075	0.49599777	− 1.68	Statistically significant at 10%		
	ATT	16.0532814	17.0.1340	− 0.9601272	0.44886877	− 2.14	Statistically significant at 5%		
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	BR	%Var	
Unmatched		0.116	26.06	0.000	39.9	33.3	59.60	570	
logpunsprw	Matched	0.001	1.23	0.873	3.2	2.2	8.71	110	

Impact on employment

Table 12 The impact of overall innovation on permanent, full-time employees

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		3.09112619	2.49780069	0.5933254	0.1153528	5.14	Statistically significant at 5%		
ATT		3.09112619	2.81030357	0.2808226	0.1956795	1.44	Statistically insignificant		
LogSample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med	B	R	%Var
av_pfemp		0.192	185.0	0.0000	24.1	Bias 14.9	119.0	0.36	100
Unmatched									
Matched		0.008	28.01	0.001	4.7	4.4	21.3	1.47	80

Table 13 The impact of overall innovation on permanent, full-time employees: production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
logav_pf emp_p	Unmatched	3.14673819	2.729499	0.4172386	0.260426	1.60	Statistically insignificant
	ATT	3.14673819	2.623096	0.523641	0.337497	1.55	Statistically insignificant
	Sample	Ps R2	LR Chi2	P > Chi2	Mean bias	Med	BR%Var
	Unmatched	0.196	51.55	0.000	34.8	Bias 12.2	113.60.1667
	Matched	0.001	1.66	0.799	3.7	4.4	8.20.71100

Table 14 The impact of overall innovation on permanent, full-time employees: non-production worker

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
logav_pfe mp_npSample	Unmatched	2.207986	1.6616793	0.54630	0.175090	3.12	Statistically significant at 5%
	ATT	2.207986	1.7648047	0.44318	0.277321	1.66	Statistically significant at 10%
	Sample	Ps R2	LR Chi2	P > Chi2	Mean bias	Med	B R %Var
	Unmatched	0.167	72.54	0.0000	31.1	Bias 19.7	104.6 0.33 100
	Matched	0.003	4.22	0.646	4.0	3.9	12.1 1.13 0

Table 15 The impact of overall innovation on permanent, full-time employees: skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
	Unmatched	2.699329	2.235992	0.463337	0.24952	1.86	Statistically significant at 10%
	ATT	2.699329	2.011102	0.688308	0.33312	2.07	Statistically significant at 5%
Sample		Ps R2	LR Chi2	P>Chi2	Mean bias	Med Bias	B R %Var
logav_pfUnmatched		0.208	56.78	0.000	48.0	39.3	122.0 0.35 100
emp_sp Matched		0.000	0.26	0.992	1.8	1.9	3.3 0.59 50

Table 16 The impact of overall innovation on permanent, full-time employees: unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
logav_pf	Unmatched	2.4521305	2.87937411	- 0.427246	0.27390	- 1.56	Statistically insignificant at 10% Statistically insignificant
	ATT	2.4521305	2.88081235	- 0.428681	0.42709	- 1.00	
emp_unpSample		Ps R2	LR Chi2	P>Chi2	Mean bias	Med bias	B R %Var
Unmatched		0.121	27.86	0.000	39.8	34.0	91.5 0.53 0
Matched		0.003	2.47	0.650	4.5	3.9	11.8 1.18 100

Appendix C

The impact of product innovation

Table 17 Impact of product innovation on productivity of labor

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	14.3998572	14.8500001	— 0.4501429	0.20619157	— 2.18	Statistically significant at 5%		
	ATT	14.3998572	15.1784527	— 0.7785955	0.20967717	— 3.71	Statistically significant at 5%		
Sample Logppw		Ps R2 0.221	LR Chi2 235.68	P > Chi2 0.000	Mean bias 29.0	Med bias 19.0	B 128.4	R 0.42	%Var 100
Unmatched									
Matched		0.006	17.26	0.045	6.2	5.6	18.0	1.40	100

Table 18 The impact of product innovation on productivity of production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	63,985,773.3	7,571,771.3	58,253,331.3	56,414,060.0	1.28	Statistically insignificant		
	ATT	63,985,773.3	12,725,160.6	57,706,509.7	51,260,612.6	3.53	Statistically significant at 5%		
Sample pprw		Ps R2 0.317	LR Chi2 105.65	P > Chi2 0.000	Mean bias 37.8	Med bias 26.3	B 155.6	R 0.22	%Var 100
Unmatched									
Matched		0.021	30.44	0.000	7.6	5.5	34.3	1.73	50

Table 19 The impact of product innovation on productivity of non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	15.782777	17.2095305	— 1.4267535	0.32235521	— 4.43	Statistically significant at 5%		
	ATT	15.782777	17.1473783	— 1.3646012	0.41868395	— 3.26	Statistically significant at 5%		
Sample logpnprw		Ps R2 0.226	LR Chi2 113.4	P > Chi2 0.000	Mean bias 31.8	Med bias 14.8	B 125.9	R 0.30	%Var 100
Unmatched									
Matched		0.015	21.15	0.002	6.7	8.7	25.9	6.9	75

Table 20 The impact of product innovation on productivity of skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		15.2350435	17.3241392	— 2.089065	0.4557104	— 4.58	Statistically significant at 5%		
ATT		15.2350435	17.6963875	— 2.461313	0.5498854	— 4.48	Statistically significant at 5%		
logpsprw	Sample	Ps R2	LR Chi2	P > Chi2	Mean bias	Med	B	R	%Var
Unmatched		0.267	88.50	0.000	50.5	Bias 25.5	141.9	0.26	100
Matched		0.005	6.65	0.156	5.1	3.6	16.6	3.31	100

Table 21 The impact of product innovation on productivity of unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
	Unmatched	16.024084	16.984268	− 0.9601836	0.42924529	− 2.24	Statistically significant at 5%
	ATT	16.024074	17.041523	− 1.0174391	0.42774269	− 2.13	Statistically significant at 5%
	Sample	Ps R2	LR Chi2	P>Chi2	Mean bias	Med bias	BR%Var
	Unmatched	0.181	50.10	0.000	46.8	29.5	113.10.400
logpunsprw	Matched	0.014	12.28	0.015	5.11	6.04	27.81.270

Impact on employment

Table 22 The impact of product innovation on permanent, full-time employees

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
Unmatched		3.1122680	2.477799	0.6344688	0.10505417	6.04	Statistically significant at 10%
ATT		3.1122680	2.825249	0.2870187	0.18634392	1.54	Statistically insignificant
LogSample	Ps R2		LR Chi2	P>Chi2	Mean bias	Med B	R %Var
av_pfemp	0.232	258.78	0.0000	26.9	Bias 17.3	132.2	0.37 100
Unmatched							
Matched		0.020	65.96	0.0000	5.10	5.30 30.2	3.76 100

Table 23 The impact of product innovation on permanent, full-time employees: production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
	Unmatched	3.1677186	2.662792	0.504925	0.2304213	2.19	Statistically significant at 5%
	ATT	3.1677186	3.002370	0.165348	0.4198013	0.39	Statistically insignificant
	Sample	Ps R2	LR Chi2	P>Chi2	Mean bias	Med	BR%Var
logav_pf emp_p	0.305	97.70	0.000	43.8	Bias 11.6	149.90.14 100	
	Unmatched						
	Matched	0.053	73.09	0.000	7.17	6.8	55.1 1.4067

Table 24 The impact of product innovation on permanent, full-time employees: non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
	Unmatched	2.218846	1.7506586	0.468188	0.156391	2.99	Statistically significant at 5%
	ATT	2.218846	2.0689408	0.149906	0.259771	0.58	Statistically insignificant
	Sample	Ps R2	LR Chi2	P>Chi2	Mean bias	Med B	R %Var
logav_pfe mp_np	0.229	118.73	0.0000	31.2	Bias 13.1	126.6	0.28 100
	Unmatched						
	Matched	0.008	12.27	0.056	4.7	3.6 20.0	1.73 25

Table 25 The impact of product innovation on permanent, full-time employees: skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	2.691379	2.409337	0.282041	0.220471	1.28	Statistically insignificant at 5%		
	ATT	2.691379	2.091493	0.599885	0.342870	1.75	Statistically significant at 10%		
Sample logav_p		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.267	88.44	0.000	48.2	25.6	142.0	0.26	100
Matched		0.028	36.73	0.000	8.18	10.2	19.39	1.08	100

Table 26 The impact of product innovation on permanent, full-time employees: unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
logav_pf	Unmatched	2.44791	2.900911	− 0.453001	0.240748	− 1.88	Statistically significant at 10%		
	ATT	2.44791	2.57467	− 0.126763	0.443313	− 0.29	Statistically insignificant		
emp_unp		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.186	52.95	0.000	47.6	30.1	115.1	0.39	0
Matched		0.012	11.59	0.021	6.9	7.6	25.8	1.30	100

Appendix D

The impact of process innovation Employment

Table 27 Impact of process innovation on productivity of permanent, full-time employees

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	14.2553	14.3064	− 0.05108	0.13012	− 0.39	Statistically insignificant at 5%		
	ATT	14.2553	15.5008	− 1.24543	0.58214	− 2.14	Statistically significant at 5%		
Sample logppw		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.114	316.55	0.000	32.3	30.8	85.2	0.98	100
Matched		0.004	18.29	0.011	3.5	2.0	14.9	1.48	100

Table 28 The impact of process innovation on productivity of production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	60,264,582.8	26,811,382.2	33,453,470.6	20,271,099.9	1.65	Statistically significant at 10%		
	ATT	60,264,852.8	20,424,237.3	39,840,615.5	20,615,448.6	1.93	Statistically significant at 10%		
Sample pprw		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.083	90.77	0.000	27.0	26.0	73.1	0.83	50
Matched		0.004	8.54	0.288	4.70	3.5	14.7	0.98	100

Table 29 The impact of process innovation on productivity of non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	15.384186	15.65905	− 0.274870	0.220232	− 1.25	Statistically insignificant at 5%		
	ATT	15.384186	16.16713	− 0.782950	0.336424	− 2.33	Statistically significant at 5%		
Sample logpnprw		Ps R2	LR Chi2	P>Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.063	7.0.93	0.0000	23.5	13.0	63.2	1.10	100
Matched		0.001	2.44	0.785	2.7	2.1	7.9	1.31	67

Table 30 The impact of process innovation on productivity of skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		14.82534	14.7137	0.111595	0.26169	0.43	Statistically insignificant at 5%		
ATT		14.82534	15.9059	− 1.08058	0.3868	− 2.79	Statistically significant at 5%		
logpsprw	Sample	Ps R2	LR Chi2	P>Chi2	Mean bias	Med	B	R	%Var
Unmatched		0.070	65.0	0.000	33.8	Bias 31.7	67.7	1.36	50
Matched		0.003	4.86	0.433	4.8	5.6	11.9	1.24	0

Table 31 The impact of process innovation on productivity of unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	15.70756	15.98927	− 0.281706	0.304120	− 0.93	Statistically insignificant at 10%		
	ATT	15.70756	16.194755	− 0.48718	0.362493	− 1.34	Statistically insignificant at 5%		
Sample logpunsprw		Ps R2	LR Chi2	P>Chi2	Mean bias	Med bias	BR	%Var	
Unmatched		0.056	34.4	0.00	24.4	25.8	61.01	0.440	
Matched		0.005	5.79	0.447	5.2	5.2	15.90	9.20	

Table 32 The impact of process innovation on permanent, full-time employees

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		3.286367	2.64532	0.64104	0.05949	10.77	Statistically significant at 10%		
ATT		3.28637	3.07825	0.20810	0.34019	0.61	Statistically insignificant		
LogSample av_pfemp		Ps R2	LR Chi2	P>Chi2	Mean bias	Med	B	R	%Var
Unmatched		0.109	335.92	0.000	30.3	Bias 26.4	83.0	0.98	100
Matched		0.001	6.55	0.477	3.1	1.9	8.3	1.18	100

Table 33 The impact of process innovation on permanent, full-time employees: production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	3.30916	2.714212	0.59495	0.11348	5.24	Statistically significant at 5%		
	ATT	3.30916	2.74037	0.56879	0.13902	4.09	Statistically significant at 5%		
logav_p ^{sample}		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
emp_p		0.034	36.19	0.000	15.8	5.6	46.9	1.11	25
Unmatched									
Matched		0.005	9.91	0.078	4.2	5.3	15.8	1.72	50

Table 34 The impact of process innovation on permanent, full-time employees: non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	2.33948	1.77788	0.56159	0.09928	5.66	Statistically significant at 5%		
	ATT	2.33948	1.86820	0.47127	0.15593	3.02	Statistically significant at 5%		
logav_pfe		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
mp_npSample		0.070	86.45	0.00	19.1	11.1	66.3	0.94	75
Unmatched									
Matched		0.001	3.45	0.751	2.3	2.0	8.8	1.70	50

Table 35 The impact of process innovation on permanent, full-time employees: skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		2.88126	2.31236	0.56889	0.1201	4.74	Statistically significant at 5%		
ATT		2.88126	2.60821	0.27304	0.14331	1.91	Statistically significant at 10%		
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med	BR%	Var	
logav_pf emp_sp		0.069	64.17	0.000	30.8	Bias	67.2	1.34	50
Unmatched		0.005	9.67	0.139	4.8	30.3	16.81	1.70	
Matched						4.3			

Table 36 The impact of process innovation on permanent, full-time employees: unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
Unmatched		2.49999	2.2293	0.27058	0.14444	1.87	Statistically significant at 10%		
ATT		2.49999	2.4163	0.08360	0.27208	0.31	Statistically insignificant		
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med	BR%	Var	
logav_pf emp_sp		0.053	35.8	0.000	31.3	Bias	59.3	1.44	50
Unmatched		0.001	2.11	0.715	3.9	27.6	9.01	2.3	0
Matched						3.9			

Appendix E

The impact of R&D innovation

See my para Tables 37, 38, 39, 40, 41

Table 37 Impact of RD innovation on productivity of permanent, full-time employees

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	14.21571	14.14337	0.072339	0.12161	0.59	Statistically insignificant at 5%		
	ATT	14.21571	14.6837	− 0.46801	0.23961	− 1.95	Statistically significant at 5%		
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
logppw	Unmatched	0.135	410.41	0.000	26.1	16.2	92.9	1.13	100
	Matched	0.004	12.64	0.049	4.1	2.5	15.4	1.35	50

Table 38 The impact of RD innovation on productivity of production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	68,237,022	34,831,090.8	33,405,931.7	16,654,732	2.01	Statistically significant at 5%		
	ATT	68,237,022	74,501,138.2	− 6,264,115.69	28,140,719.1	− 0.22	Statistically insignificant at 5%		
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med	B	R	%Var
pprw	Unmatched	0.092	131.71	0.000	17.3	8.2	75.0	1.25	50
	Matched	0.002	3.55	0.738	2.4	1.5	10.9	3.19	50

Table 39 The impact of RD innovation on productivity of non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	15.3966	15.5650	− 0.1683	0.1866	− 0.90	Statistically insignificant at 5%		
	ATT	15.3966	15.9432	− 0.5465	0.33724	− 1.65	Statistically significant at 10%		
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
lognpw	Unmatched	0.120	173.58	0.000	20.1	6.0	86.7	1.07	25
	Matched	0.001	0.91	0.989	2.1	1.8	5.8	0.96	25

Table 40 The impact of RD innovation on productivity of skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	14.79334	14.848	− 0.0551	0.21112	− 0.26	Statistically insignificant at 5%		
	ATT	14.7933	15.50	− 0.70892	0.3779	− 1.88	Statistically significant at 10%		
logpsprw	Sample	Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B	R	%Var
Unmatched		0.097	127.75	0.00	27.7	18.8	77.1	1.27	0
	Matched	0.001	0.75	0.945	1.2	0.0	5.4	1.16	0

Table 41 The impact of RD innovation on productivity of unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
Unmatched		15.6785	15.9026	− 0.22408	0.24785	− 0.90	Statistically insignificant at 10%
ATT		15.6785	16.437	− 0.75934	0.4259	− 1.78	Statistically insignificant at 5%
Sample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med	BR%Var
logpunsprw		0.076	63.79	0.000	23.7	bias	67.3 1.52 0
Unmatched		0.000	0.35	0.986	1.60	16.8	4.70.96 0
Matched						1.0	

Employment

Table 42 The impact of RD innovation on permanent, full-time employees

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
Unmatched		3.32168	2.9046	0.417	0.0554	7.52	Statistically significant at 10%
ATT		3.32168	3.1447	0.176914	0.12858	1.38	Statistically insignificant
LogSample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med	B R %Var
av_pfpemp		0.131	425.33	0.00	17.5	bias	91.6 1.14 80
Unmatched						6.4	
Matched		0.003	9.64	0.291	3.2	2.2 12.4	1.20 60

Table 43 The impact of RD innovation on permanent, full-time employees: production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
Unmatched		3.33239	3.002	0.33031	0.09507	3.47	Close statistically significant at 5%
ATT		3.3323	2.995	0.3373	0.1343	2.51	Close statistically significant at 5%
logav_pfsample		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B R %Var
emp_p		0.087	121.10	0.000	16.8	1.3	73.0 1.24 0.0
Unmatched							
Matched		0.005	6.96	0.224	4.9	4.6	16.0 1.08 0.0

Table 44 The impact of RD innovation on permanent, full-time employees: non-production workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision
Unmatched		2.3819	2.0244	0.35748	0.0838	4.26	Statistically significant at 5%
ATT		2.38196	1.8554	0.5265	0.15211	3.46	Statistically significant at 5%
logav_pfe		Ps R2	LR Chi2	P > Chi2	Mean bias	Med bias	B R %Var
mp_npSample		0.120	190.93	0.000	17.7	5.2	86.6 1.07 20
Unmatched							
Matched		0.001	1.48	0.983	2.0	2.3	7.0 0.90 40

Table 45 The impact of RD innovation on permanent, full-time employees: skilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
	Unmatched	2.875665	2.5631	0.31255	0.09562	3.27	Statistically significant at 5%		
	ATT	2.8756	2.4085	0.46715	0.14253	3.28	Statistically significant at 5%		
Sample		Ps R2	LR Chi2	P>Chi2	Mean bias	Med bias	B	R	%Var
logav_pf		0.094	123.4	0.000	25.6	14.8	75.8	1.25	0
emp_sp	Unmatched								
	Matched	0.000	0.05	1.00	0.4	0.1	1.4	1.10	0

Table 46 The impact of RD innovation on permanent, full-time employees: unskilled workers

Variable	Sample	Treated	Controls	Difference	S. E	T-Stat	Decision		
logav_pf	Unmatched	2.5564	2.3374	0.2189	0.1178	1.86	Statistically significant at 10%		
	ATT	2.55	2.1116	0.44480	0.23167	1.92	Statistically significant at 10%		
emp_unp	Sample	Ps R2	LR Chi2	P>Chi2	Mean bias	Med bias	B	R	%Var
	Unmatched	0.075	68.77	0.0000	23.2	16.6	66.8	1.47	0.0
	Matched	0.0000	0.01	1.000	0.2	0.2	0.7	1.0	0

Appendix F

Results of the dose–response estimation (Table 47)

Table 47 (continued)

Explanatory Variables	logppw		logppr		logpnprw		logpsprw		logpunsprw	
	Coef	p-	Coef	p-	Coef	p-	Coef	p-	Coef	p-
Innovation dummy										
_lquace_int_1	1.1121	0.000								
logTotalmainproduct	0.3227	0.055					1.120	0.013	0.311	0.552
private_dshar	− 0.6910	0.056	− 0.5764	0.353	− 1.1449	0.109				
private_fshar	0.9347	0.020	0.2587	0.698	− 0.318429	0.672				
gov_shar	− 1.406	0.217	− 2.573	0.077	− 3.6185	0.014				
_ws_firmsize	0.1302	0.647	0.3269	0.570	0.21119	0.730	0.0222	0.000	0.0239	0.001
Tw_1	− 0.8315	0.013	− 0.5849	0.307	− 0.88596	0.131	− 2.24	0.000	− 1.472	0.034
Tw_2	0.1093	0.134	0.07521	0.520	0.15301	0.199	0.360	0.003	0.2207	0.142
Tw_3	− 0.0054	0.257	− 0.0039	0.589	− 0.0095	0.194	− 0.019	0.012	− 0.0108	0.272
_cons	14.72	0.000	18.8567	0.000	19.753	0.000	16.38	0.000	17.142	0.000
R2	0.1001		0.1784		0.1794		0.1232		0.0979	
F(.) p-value	13.43	9.65		9.08		19.5(0.00)		7.39(0.00)		
	(0.00)	(0.00)		(0.00)						

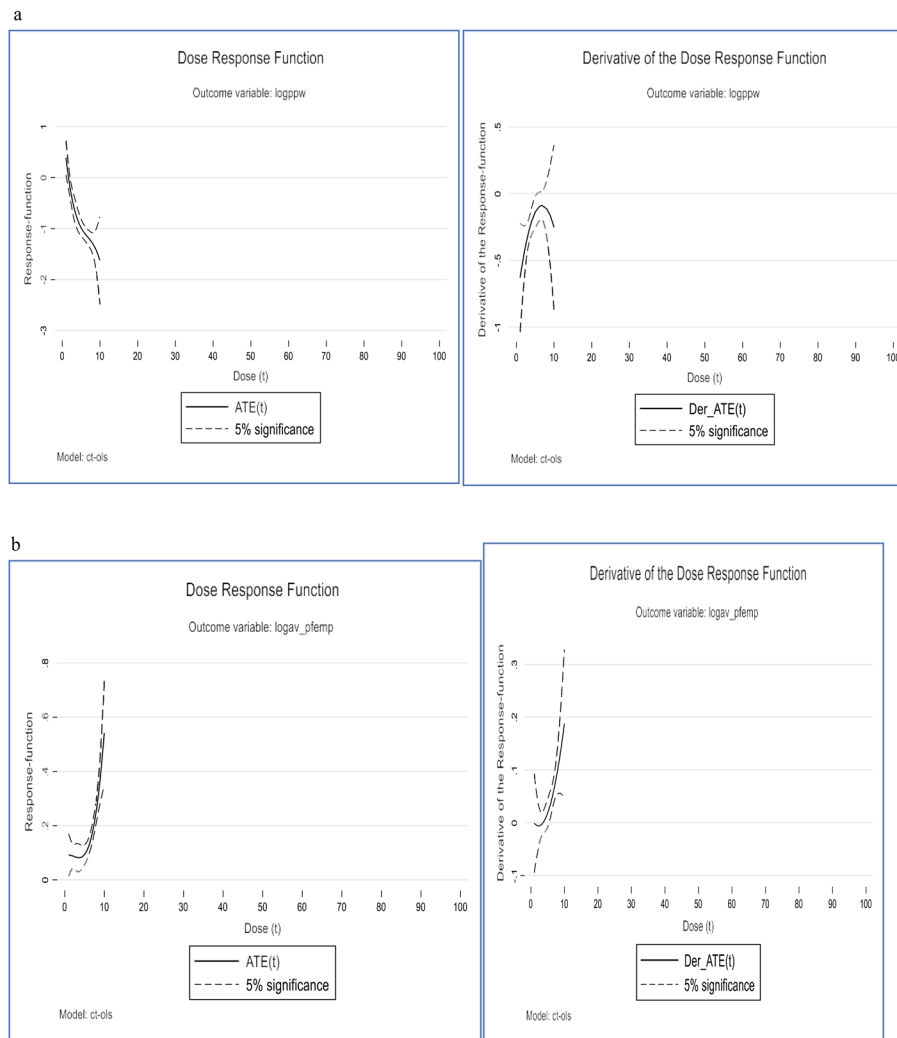


Fig. 1 Results of the dose–response estimation. **a** Overall innovation and productivity of full-time permanent employees. **b** Overall innovation and employment of full-time permanent workers. **c** Overall innovation and productivity of production workers. **d** Overall innovation and employment of production workers. **e** Overall innovation and productivity of non-production workers. **f** Overall innovation and employment of non-production workers. **g** Overall innovation and productivity of skilled production workers. **h** Overall innovation and employment of skilled production workers. **i** Overall innovation and productivity of unskilled production workers. **j** Overall innovation and employment of unskilled production workers

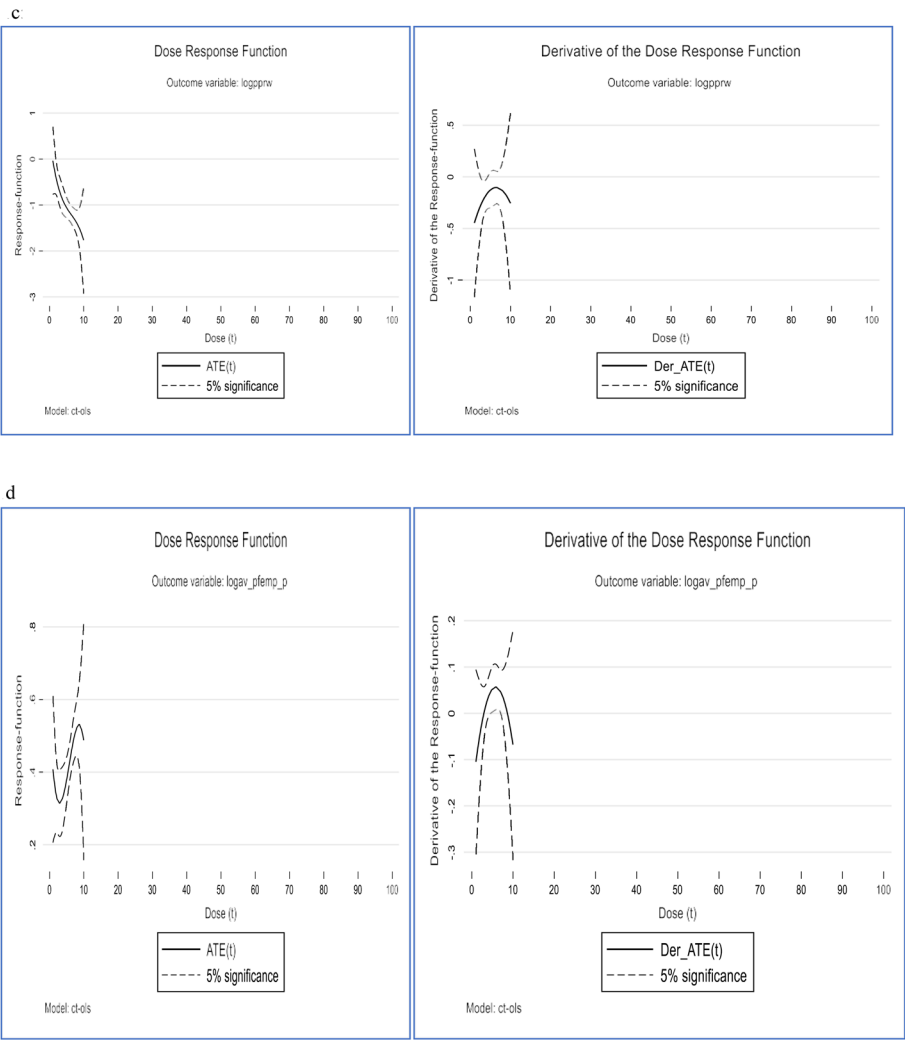


Fig. 1 continued

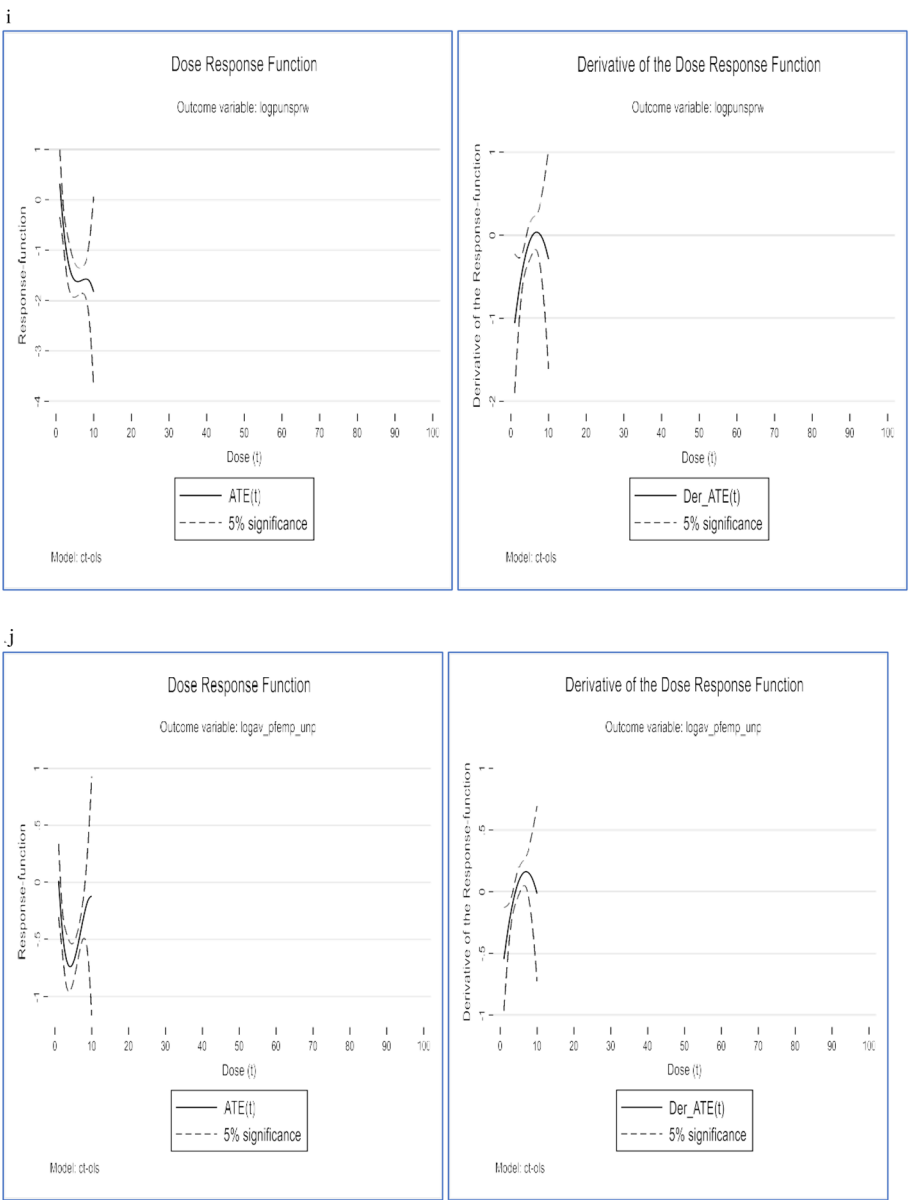


Fig. 1 continued

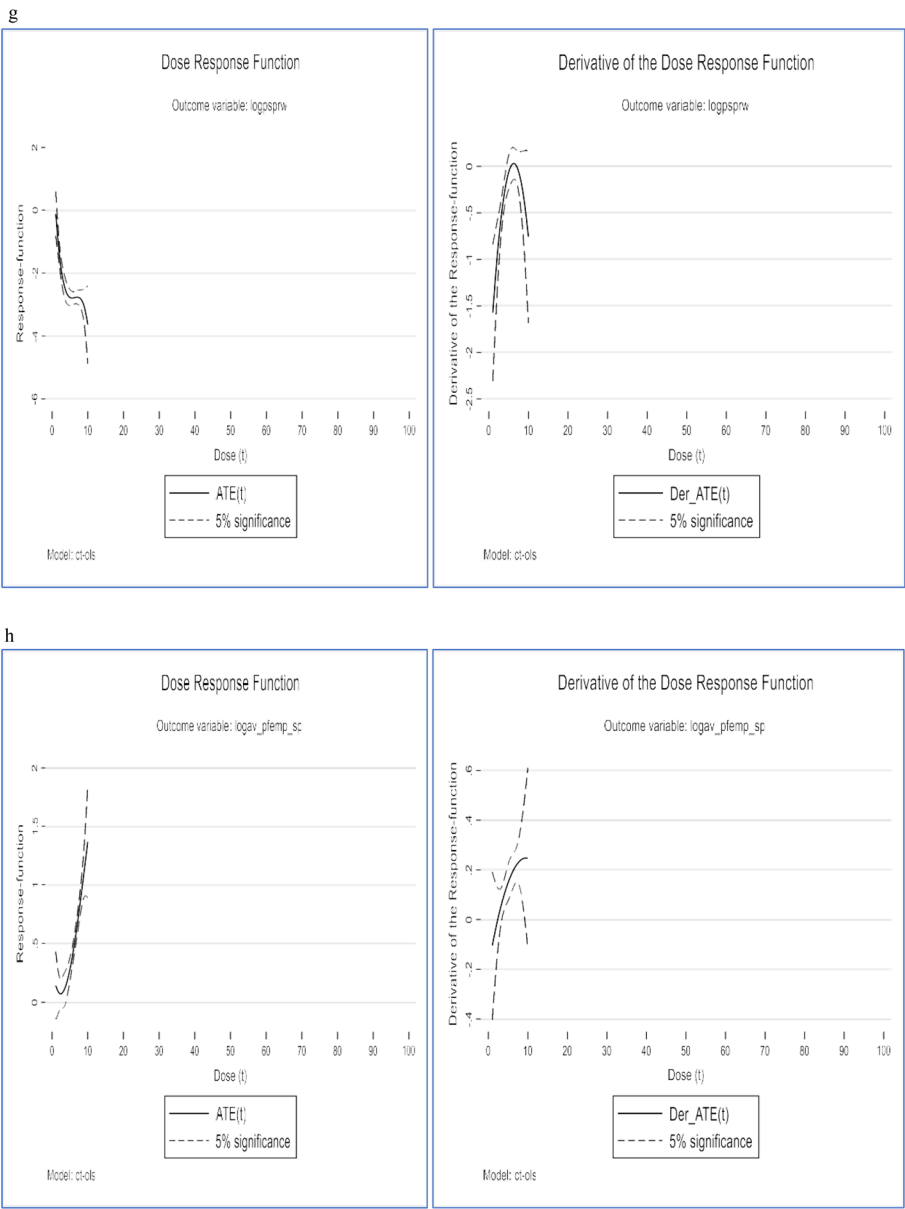


Fig. 1 continued

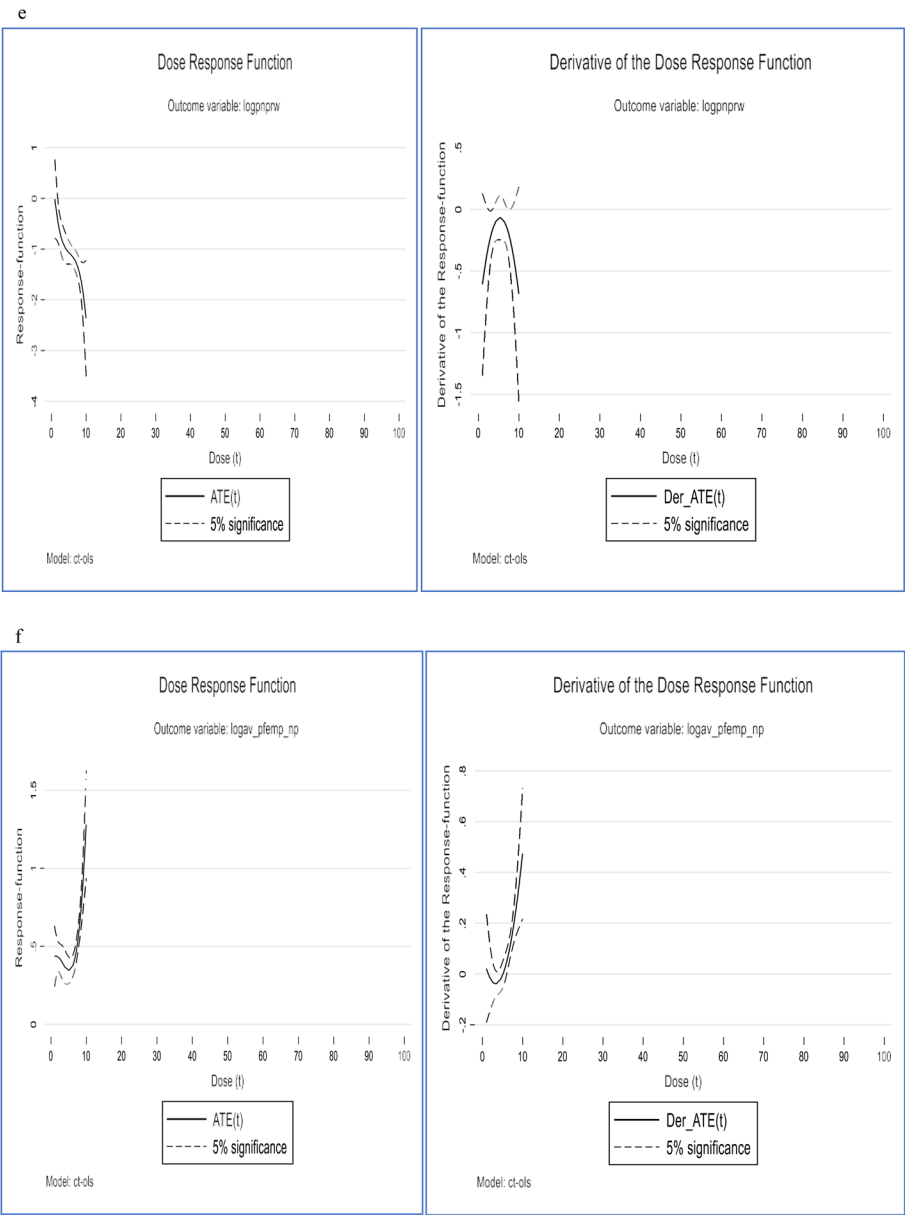


Fig. 1 continued

Acknowledgements

This study was carried out for the Program of Accompanying Research for Agricultural Innovation (PARI) as part of the cooperation between the Center for Development Research (ZEF) and AKADEMIYA2063. We are grateful to the German Federal Ministry for Economic Cooperation and Development (BMZ) for funding this work through PARI as part of the German Government's One World, No Hunger Initiative (SEWOH).

Author contributions

All authors whose names appear on the submission made substantial contributions to the conception or design of the work; or the acquisition, analysis, or interpretation of data; or the creation of new software used in the work; drafted the work or revised it critically for important intellectual content; approved the version to be published; and agree to be accountable for all aspects of the work in ensuring that questions related to the accuracy or integrity of any part of the work are appropriately investigated and resolved.

Data availability

This research goes through explaining the impact of different types of innovation on labor productivity and employment. The survey data are freely available from the World Bank Enterprise Survey. These compiled data will be made available upon reasonable request for academic use. And, the datasets generated during and/or analyzed during the current study are available from the corresponding author upon reasonable request.

Declarations

Consent for publication

The Journal and Publisher assume all authors agreed with the content that all gave explicit consent to submit and that they obtained consent for publication.

Competing interests

On behalf of all authors, the corresponding author states that there is no competing of interest.

Received: 5 August 2024 Accepted: 28 October 2024

Published online: 28 January 2025

References

- Adelino, M., Song, M., & David, R. (2017). Firm age, investment opportunities, and job creation. *Journal of Finance*. <https://doi.org/10.1111/jofi.12495>
- Adorno, V., Bernini, C., & Pellegrini, G. (2007). The impact of capital subsidies: New estimations under continuous treatment. *Giornale Degli Economisti Annali Di Economia*, 66(1), 67–92.
- Anneleen, V., Istvan, V., Mauro, V., & Lukas, V. (2022). *The Possible Implications of the Green Transition for the EU Labour Market*. ISSN 2443–8022 (online), European Economy, Discussion Paper 176.
- Armbruster, H., Bikfalvi, A., Kinkel, S., & Lay, G. (2008). Organizational innovation: The challenge of measuring non-technical innovation in large-scale surveys. *Technovation*, 28(10), 644–657.
- Atoko, K., & Tlangelani, M. (2024). Innovation and economic growth: An empirical analysis for African countries. *African Journal of Science, Technology, Innovation and Development*. <https://doi.org/10.1080/20421338.2024.2382612>
- Avenyo, E., & Konte, M. & Mohnen, P. (2019). *The employment impact of product innovations in sub-Saharan Africa: Firm-level evidence*. MERIT Working Papers 2019–019, United Nations University - Maastricht Economic and Social Research Institute on Innovation and Technology.
- Balasubramanian, N., & Sivadasan, J. (2011). What happens when firms patent? New evidence from U.S. economic census data. *Review of Economics and Statistics*, 93(1), 126–146.
- Bia, M., Flores, C. A. F., Flores-Lagunes, A., & Mattei, A. (2014). A Stata package for the application of semiparametric estimators of dose-response functions. *The Stata Journal*, 14(3), 580–604.
- Bia, M., & Mattei, A. (2008). A Stata package for the estimation of the dose-response function through adjustment for the generalized propensity score. *Stata Journal*, 8(3), 354–373.
- Blanchflower, D., & Burgess, G. (1996). Job creation and job destruction in Great Britain in the 1980s. *Industrial and Labor Relations Review*, 50, 17–38.
- Bloom, N., & Van Reenen, J. (2002). Patents, real options and firm performance. *Economic Journal*, 112(478), 97.
- Bloom, N., Van Reenen, J., & Williams, H. (2019). A toolkit of policies to promote innovation. *Journal of Economic Perspectives*, 33(3), 163–184.
- Bruno, S., & Diana, M. (2019). Innovations for sustainable development: Moving toward a sustainable future. *Journal of Cleaner Production*, 208, 325–332. <https://doi.org/10.1016/j.jclepro.2018.09.244>
- Caliendo, M., & Kopeining, S. (2005). *Some practical guidance for the implementation of propensity score matching*. IZA Discussion Paper No. 1588.
- Chen, Z., Zhiyi, Q., & Fengjun, L. (2021). Does innovation stimulate employment? Evidence from China. *Economic Modelling*, 94, 1007–1017.
- Chenery, H., & Taylor, L. (1968). Development patterns among countries and over time. *Review of Economics and Statistics*, 50, 391–416.
- Chudnovsky, D., Andrés, L., & Germán, P. (2006). Innovation and productivity in developing countries: A study of Argentine manufacturing firms' behavior (1992–2001). *Research Policy*, 35(2), 266–288.
- Cirera, X., & Sabetti, L. (2016). *The effects of innovation on employment in developing countries: Evidence from enterprise surveys*. Policy Research Working Paper. 7775, World Bank Group

- Cirera, X., Lage, F., & Sabetti, L. (2016). *ICT Use, Innovation, and Productivity: Evidence from Sub-Saharan Africa*. World Bank Policy Research, Working Paper No. 7868.
- Cohen, W., & Levinthal, D. (1989). Innovation and learning: The two faces of R&D. *The Economic Journal*, 99(397), 569–596.
- Crepon, B., Duguet, E., & Mairesse, J. (1998). Research, innovation and productivity: An econometric analysis at the firm level. *Economics of Innovation and New Technology*, 7(2), 115–158.
- Crespi, G., Tacsir, E., & Vargas, F. (2016). Innovation dynamics and productivity: Evidence for Latin America. In M. Grazi & C. Pietrobelli (Eds.), *Inter-American Development Bank, firm innovation and productivity in Latin America and the Caribbean*. Palgrave Macmillan. https://doi.org/10.1057/978-1-349-58151-1_2
- Doris, A. (2016). *Employment effects of innovation in developing countries: A summary*. Swiss Programme for Research on Global Issues for Development, Working Paper 2016/2.
- Duguet, E. (2006). Innovation height, spillovers and TFP growth at the firm level: Evidence from French manufacturing. *Economics of Innovation and New Technology*, 15(4–5), 415–442.
- Edquist, C., Hommen, L., & McKelvey, M. (2001). *Innovation and employment: Process versus product innovation* (p. 214). Edward Elgar Publishing.
- EIC. (2020). *Economic Report on Africa, governing development in Africa, the role of the State in economic transformation*. Economic Commission for Africa.
- Fagerberg, J. (2006). Innovation, technology and the global knowledge economy: Challenges for future growth. In J. Fagerberg (Ed.), *Green roads to growth. Project and conference*. University of Oslo.
- Fagerberg, J., Srholec, M., & Verspagen, B. (2010). Innovation and economic development. *Handbook of the Economics of Innovation*, 2, 833–872.
- Fagerberg, J., & Verspagen, B. (2002). Technology-gaps, innovation diffusion, and transformation: An evolutionary interpretation. *Research Policy*, 31(8–9), 1291–1304. [https://doi.org/10.1016/S0048-7333\(02\)00064-1](https://doi.org/10.1016/S0048-7333(02)00064-1)
- Farrell, M. (1957). The measurement of productive efficiency. *Journal of the Royal Statistical Society Series*, 3, 253–290.
- Fendoğlu, E., & Polat, M. (2021). The relationship between R&D expenditures and economic growth: Panel data analysis in selected new industrializing countries. *Kent Akademisi*, 14(3), 728–747. <https://doi.org/10.35674/kent.976698>
- Foss, N. (1998). *The Theory of the Firm: An Introduction to Themes and Contributions*. Institut for Industriøkonomi og Virksomhedsstrategi, Handelshøjskolen i København. Working Paper/Department of Industrial Economics and Strategy. Copenhagen Business School No. 1998–7.
- Foster, L., Haltiwanger, F., & Krizan, C. (2001). Aggregate productivity growth: Lessons from microeconomic evidence. In C. R. Hulten, E. R. Dean, & M. J. Harper (Eds.), *New developments in productivity analysis* (pp. 303–363). University of Chicago Press.
- Foxon, T. (2003). *Inducing innovation for a low-carbon future: Drivers, barriers, and policies—a report for the carbon trust*. The Carbon Trust.
- Freeman, C., & Soete, L. (1987). *Technical change and full employment*. Basil Blackwell.
- Getaw, T., Florence, G., & Tendai, G. (2022). *Firm-Level Innovation and Business Obstacles in Sub-Saharan Africa: A cross-country analysis*. Working Paper, AKADEMIYA2063.
- Giorgos, M., & Manolis, M. (2021). History, knowledge, and sustainable economic development: The contribution of John Stuart Mill's grand stage theory. *Sustainability*, 2021(13), 1468. <https://doi.org/10.3390/su13031468>
- Goldman, S., Nagel, R., & Preiss, K. (1995). *Agile competitors and virtual organizations: Strategies for enriching the customer*. Van Nostrand Reinhold.
- Greenan, N. (2003). Organizational change, technology, employment, and skills: An empirical study of French manufacturing. *Cambridge Journal of Economics*, 27(2), 287–316.
- Griffith, R., Redding, S., & Simpson, H. (2004). Foreign ownership and productivity: New evidence from the service sector and the R&D lab. *Oxford Review of Economic Policy*, 20, 3.
- Guillermo, A., Andrés, B., & Joost, H. (2020). The effect of innovation on skilled and unskilled workers during bad times. *Journal of Structural Change and Economic Dynamics*, 52, 141–158.
- Hall, B. (2011). Innovation and productivity. *Nordic Economic Policy*, 2, 167–204.
- Hall, B., Francesca, L., & Jacques, M. (2009). Innovation and productivity in SMEs: Empirical evidence for Italy. *Small Business Economics*, 33(1), 13–33.
- Hameed, T., Alemayehu, F. K., & Kumbhakar, S. C. (2024). Green innovation in Norwegian firms: Navigating the complexity of productivity and performance. *Technological Forecasting and Social Change*, 209, 123786.
- Hanan, M. & Amira E. (2020). *Innovation and productivity in developing economies*. Working Paper Series, No 340, African Development Bank Group. <https://www.researchgate.net/publication/342468254>
- Harrison, R., Jaumandreu, J., Mairesse, J., & Peters, B. (2014). Does innovation stimulate employment? A firm-level analysis using comparable micro-data from four European countries. *International Journal of Industrial Organization*, 35, 29–43.
- He, Z., Chen, Z., & Feng, X. (2023). The role of green technology innovation on employment: does industrial structure optimization and air quality matter? *Environmental Sciences Europe*, 35, 59. <https://doi.org/10.1186/s12302-023-00773-w>
- Heckman, J., Ichimura, H., & Todd, P. (1997). Matching as an econometric evaluation estimator: Evidence from evaluating a job training programme. *Review of Economic Studies*, 64(4), 605–654.
- Hirano, K., & Imbens, G. (2004). The propensity score with continuous treatments. *Applied Bayesian modeling and causal inference from incomplete-data perspectives: An essential journey with Donald Rubin's statistical family* (pp. 73–84). Wiley.
- Imai, K., & Dyk, D. (2004). Causal inference with general treatment regime: Generalizing the propensity score. *J. Amer. Statist. Assoc.*, 99, 854–866.
- Imbens, G. (2000). The role of the propensity score in estimating dose-response functions. *Biometrika*, 87, 706–710.
- Jabir, A., & Nadia, Y. (2021). International quality certification and business performance of Indian firms: Evidence from enterprise survey data. *Global Business Review, International Management Institute*, 22(6), 1459–1470.
- Janz, N., Lööf, H., & Peters, A. (2004). Firm-level innovation and productivity: Is there a common story across countries? *Problems and Perspectives in Management*, 2, 1–22.
- Jorgenson, D. (2011). Innovation and productivity growth. *American Journal of Agricultural Economics*, 93(2), 276–296.

- Katsoulacos, Y. (1986). *The employment effect of technical change*. Brighton: Wheatsheaf.
- Kongsamut P., Rebelo S., and Xie D. (1997). *Beyond Balanced Growth*. NBER Working Paper Series 6159, Cambridge, MA 02318
- Kuznets, S. (1966). *Modern economic growth*. Yale University Press.
- Lechner, M. (2001). Identification and estimation of causal effects of multiple treatments under the conditional independence assumption. In M. Lechner & F. Pfeiffer (Eds.), *Econometric evaluation of labour market policies* (pp. 43–58). Physica.
- Mairesse, J., & Sassenou, M. (1991). *R&D Productivity: A Survey of Econometric Studies at the Firm Level*. NBER Working Paper No. w3666.
- Mairesse, J. & Robin S. (2009). *Innovation and productivity in France: a firm level analysis for Manufacturing and Services using CIS3 and CIS4 data (1998–2000 and 2002–2004)*. Paris: Crest-Ensaie 7 (2009)
- Mairesse, J. and Mohnen, P. (2010). *Using Innovations Surveys for Econometric Analysis*. No 15857. NBER Working Papers from National Bureau of Economic Research, Inc.
- Mairesse, J., Mohnen, P., & Kremp, E. (2005). The importance of R&D and innovation for productivity: A reexamination in Light of the 2000 French Innovation Survey. *Annales D'économie Et De Statistique*, 79(80), 489–529.
- Marine, G. (2017). Pseudo-panel methods and an example of application to Household Wealth data. *Economie et Statistique / Economics and Statistics*, Institut National de la Statistique et des Etudes Economiques (INSEE), issue 491–492, pp. 109–130.
- Masso, J. & Vahter, P. (2008). *Technological innovation and productivity in late-transition Estonia: Econometric evidence from innovation surveys*. The University of Tartu Faculty of Economics and Business Administration Working Paper No. 61–2008
- Mebratie, A. & Bedi, A. (2013). *Foreign direct investment, black economic and Services, 1998–2000 and 2002–2004*, mimeo, CREST-ENSAE.
- Mohnen, P., & Hall, B. (2013). Innovation and productivity: An update. *Eurasian Business Review*, 3(1), 47–65.
- Naveed, A., & Wang, C. (2023). Innovation and labor productivity growth moderated by structural change: Analysis in a global perspective. *Technovation*, 119, 102554.
- Nemet, G. (2007). *Policy and innovation in low-carbon energy technologies*. Dissertation abstracts international, 68/08.
- OECD (2005). *Proposed guidelines for collecting and interpreting technological innovation data: Oslo Manual*.
- Okumu, I., Bbaale, E., & Guloba, M. (2019). Innovation, and employment growth: evidence from manufacturing firms in Africa. *Journal of Innovation and Entrepreneurship*, 8, 7. <https://doi.org/10.1186/s13731-019-0102-2>
- Peters, B., Hud, M., Dachs, B., & Köhler, C. (2017). *Employment effects of innovations over the business cycle: Firm-level evidence from European countries* (No. A12-V3). <http://hdl.handle.net/10419/168211>
- Philip, G, Robert, G., & Jamie, S. (2012). *Innovation theory: A review of the literature*. Imperial College Center for Energy Policy and Technology, ICEPT Working Paper.
- Pianta, M. (2005). Innovation and employment. In J. Fagerberg, D. Mowery, & R. Nelson (Eds.), *The Oxford Handbook of Innovation* (pp. 568–597). Oxford University Press.
- Pierre, M., & Hall, B. (2013). *Innovation and productivity: An update*. UNU-MERIT working papers, ISSN 1871–9872
- Raffo, J., Stéphane, L., & Egidio, L. (2008). Northern and Southern Innovative: A comparison across European and Latin American Countries. *European Journal of Development Research*, 20(2), 219–239.
- Rasi, K., & Ester, M. (2016). Towards green growth: How does green innovation affect employment? *Research Policy*, 45(6), 1218–1232.
- Roper, S., Du, J., & Love, J. H. (2008). Modeling the innovation value chain. *Research Policy*, 37(6–7), 961–977.
- Rosenbaum, P., & Rubin, D. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika*, 70(1), 41–55.
- Ruttan, V. (2001). *Technology, growth and development: an induced innovation perspective*. Oxford University Press.
- Schumpeter, J. (1934). *The theory of economic development*. Harvard University Press.
- Simanis, E., & Hart, S. (2009). Innovation from the inside out. *MIT Sloan management review*.
- Simpkin, V., Namubiru, L. C., & Mossialos, E. (2019). Investing in Health R&D: Where we are, what limits us, and how to make progress in Africa. *BMJ Glob of Health*, 2019(4), e001047. <https://doi.org/10.1136/bmjgh2018-001047>
- Smolny, W. (1998). Innovations, prices, and employment: A theoretical model and an empirical application for West German manufacturing firms. *Journal of Industrial Economics*, 46(3), 359–381.
- Spiezia, V., & Vivarelli, M. (2002). Innovation and employment: A critical survey. *Book Chapter*. <https://doi.org/10.7551/mitpress/5609.003.0009>
- Syverson, C. (2011). What determines productivity? *Journal of Economic Literature*, 49(2), 326–365.
- Tomasz, K., & Arkadiusz, K. (2018). Is innovation the key to solving the productivity paradox? *Empirical Paper, Journal of Innovation & Knowledge*. <https://doi.org/10.1016/j.jik.2017.12.010>
- Twumasi, B., Quartey, P., & Adu-Danso, E. (2022). The effects of innovation on the quantity and quality of jobs: Evidence from sub-Saharan Africa. *Innovation and Development*, 14(1), 165–187. <https://doi.org/10.1080/2157930X.2022.2116784>
- UNESCO (2018). *Skills and innovation in G20 countries*. Information paper No. 50, June 2018, UIS/2018/STI/IP/50.
- Van Leeuwen, G., & Klomp, L. (2006a). On the contribution of innovation to multi-factor productivity growth. *Economics of Innovation and New Technology*, 15(4–5), 367–390.
- Van Leeuwen, G., & Klomp, L. (2006b). On the contribution of innovation to multifactor productivity growth. *Economics of Innovation and New Technology*, 15(4/5), 367–390.
- Vivarelli, M. (1995). *The economics of technology and employment: Theory and empirical evidence*. Edward Elgar Publishing.
- Vivarelli, M. (2007). Innovation, and employment: A Survey. *SSRN Journal*. <https://doi.org/10.2139/ssrn.969621>
- Vivarelli, M. (2014). Innovation, employment, and skills in advanced and developing countries: A survey of the economic literature. *Journal of Economic Issues*, 48(1), 123–154.
- Vivarelli, M. (2015). *Innovation and employment*, IZA World of Labor, ISSN 2054–9571. Institute for the Study of Labor (IZA). <https://doi.org/10.15185/izawol.154>

- Vollenbroek, F. (2001). Sustainable development and the challenge of innovation. *Journal of Cleaner Production*, 10, 215–223.
- WB,. (2019). *World development indicator report*. Washington, DC: Statistical data online version, World Bank.

Publisher's Note

Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.